

AN ARCHIVAL ALMA SEARCH FOR UNRESOLVED SPECTRAL-CARRIER TECHNOSIGNATURES
TOWARD STARS WITHIN 40 PC AT 114–873 GHzGLENN J. WHITE^{1,2} AND ROBIN DEY³

ABSTRACT

No star within 40 pc has been searched above about 30 GHz for a narrowband transmitter. The ALMA archive reaches it incidentally: dust-disc and planet-formation observations record a spectrum of the star at every field centre. Re-reduced uniformly from the raw visibilities, they are searched for a drifting, unresolved carrier at the stellar position in 402 fine-channel windows toward 60 systems, beside a coarse-channel bound discriminating no carrier. No event met the confirmation criterion of recurrence. Of the 56 threshold crossings, four arise in one execution block failing a quality criterion devised after it was seen, and many of the rest coincide with a catalogued molecular transition, among them β Pictoris’s carbon monoxide, recovered blind. Each unattributed crossing was tested at the predicted stellar-frame channel and across every later covering window and drift; none returned, including the two that lead their own control fields. That excludes a carrier holding its frequency inside the window searched and radiating through both observations, not a persistent transmitter of any kind; the remainder is consistent with chance, to the width the control positions allow. Confirmation was possible for 39 of the 60 systems and impossible at any power for the 21 observed once or within a single day. Depth is a recovery fraction through the single-epoch trigger and not a threshold: injected carriers come back nine times in ten at 1.6×10^{15} W for the median system. Molecular-line confusion and sparse repeat coverage, not sensitivity, limit a search of this kind.

Subject headings: extraterrestrial intelligence, radio lines: stars, submillimetre: stars, surveys, astrobiology

1. INTRODUCTION

The public ALMA archive holds thousands of hours of millimetre and submillimetre observations taken for other reasons, mostly studies of dust discs and planet formation. Their visibilities allow a spectrum at the position of the star at the centre of each field, at a resolution and depth set by the original observer’s correlator setup. This paper searches those spectra for a narrowband transmitter at the star’s own position. Nothing of the kind survived.

Searches for *technosignatures* — detectable imprints of technology on the electromagnetic spectrum, intended or not (Sheikh 2020; Vidal et al. 2026) — have examined many thousands of stars over six decades, yet the last published accounting of that effort covers only 6×10^{-18} of the space of transmitter position, power, frequency, bandwidth, polarisation and repetition rate (Wright et al. 2018), and essentially all of it lies below ~ 30 GHz. No published search above that frequency has reached a star within 40 pc, though more distant stars have been searched in the millimetre band (Mason et al. 2025). The band is attractive because a carrier arrives less distorted and a given aperture delivers more gain per watt supplied; what makes it hard belongs to the archive rather than to the sky, and §2 sets that out.

The signal sought is an *unresolved spectral carrier*: emission confined to a few channels, unresolved at the stellar position, and present at more than one epoch.

Each qualifier is a restriction, and the last is the strictest: it is recurrence rather than brightness that separates an interesting event from a detection (Sheikh et al. 2021). The archive holds a second, separated epoch for some of these systems and not for others, so what could be searched and what could be confirmed are different extents. The frequency is allowed to drift, as it must for a transmitter carried on a rotating, orbiting body (§4.1). “Narrowband” here describes the assumed transmitter and never the resolving power of the archived channels. The targets were chosen by other programmes for other purposes, so every statistical statement that follows is about an archive-selected sample (§6.3).

2. BACKGROUND

Radio searches for technosignatures began at centimetre wavelengths and have largely stayed there. The first targeted experiments (Cocconi & Morrison 1959; Drake 1961), the programmes reviewed by Tarter (2001) and Project Phoenix (Backus & Project Phoenix Team 2002) all worked below a few gigahertz. The modern generation divides into targeted and commensal surveys at centimetre and metre wavelengths (Zhang et al. 2020; Czech et al. 2021; Li et al. 2026; Johnson et al. 2023; Choza et al. 2024), several of them riding back-ends built for other science (Morrison et al. 2023), with wide-field low-frequency coverage from the MWA (Tremblay & Tinney 2024) and single-dish C- and K-band work at 6 and 18 GHz with the Sardinia Radio Telescope (Manunza et al. 2025). Machine-learning interference rejection is now central to that effort (Ma et al. 2023; Jacobson-Bell et al. 2025; Pardo et al. 2025). The classical threshold search used here lacks that capability and instead uses a spatial test on the star against control positions in the same

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field, which a single dish cannot make. A narrowband search also reaches only one class of signal: broadband planetary-scale leakage (Garrett 2026) is outside it by construction.

Above about 30 GHz the record is nearly empty, and that is what makes this frequency range new. ALMA’s 10 receiver bands cover 35 to 950 GHz, and one of them has been searched for technosignatures. The lowest, 35–50 GHz, does hold observations of 4 stars of this sample, but all from one programme, all in a continuum mode whose 31.25 MHz channels are 2048 times coarser than the narrowest channel searched here, and all proprietary until 2027: the coverage exists and cannot carry a carrier search. The band above it holds none at all. For continuity with the centimetre literature, the Arecibo planetary radar radiated 2.0×10^{13} W at 2.38 GHz (Sheikh et al. 2025); an aperture of that diameter is not realisable in the millimetre band, so the illustrative benchmark of §6.3 is instead a dish of 12 m. The limits of this paper are stated as EIRP and need no transmitter model.

Mason et al. (2025) is the only prior millimetre search and the only comparable archival one. They searched Band 3 windows toward 28 Gaia DR3 stars lying in the fields of 4 ALMA calibrators, observed for other science and so usable for free. Their nearest target is at 1.01 kpc, where a 5σ flux threshold corresponds to $\text{EIRP}_{\min} > 6.9 \times 10^{17}$ W. That target lies between 25 and 277 times further away than the stars searched here, so at a fixed received flux an EIRP ratio between the two surveys would be 3–5 orders of magnitude of d^2 and almost nothing else. The quantity that does not depend on distance is the minimum detectable *received* flux, $F_{\min} = S_{\min} \delta\nu$, and it has to be compared at the channel width each experiment recorded: $S_{\min}$ falls as $\delta\nu^{-1/2}$ while $F_{\min}$ rises as $\delta\nu^{1/2}$, so putting a coarse window on a fine channel arithmetically credits it with a depth it never reached. Mason et al. reach 3.8×10^{-23} – 5.8×10^{-23} W m $^{-2}$ across their four fields, on a 30.52 kHz channel at a bare 5σ flux threshold. On that same threshold convention and on native channels throughout, the windows searched here reach 4.8×10^{-24} W m $^{-2}$ at best — a factor 10 deeper, in a channel 15.3 kHz wide — and 3.0×10^{-22} W m $^{-2}$ at the median, a factor 6 shallower in a channel 488 kHz wide. Per channel this survey is therefore the shallower of the two except in its finest windows, and its advantage lies elsewhere: 60 stellar systems within 40 pc rather than distant stars in four calibrator fields, and a frequency range reaching about nine times higher. Mason et al. also identified the obstacles that shape the present method: drift rates scale with frequency, smearing erodes sensitivity unless drift is searched for, and the molecular line forest raises the confusion background. Every window reported here was re-reduced and re-searched from the raw archive with the pipeline of §4.

Figure 1(a) places this survey among published searches, in minimum detectable EIRP against distance. For a carrier narrower than every programme’s channel that limit is a total radiated power rather than a spectral power density, so the axis is commensurable; what differs is the threshold convention, and both are drawn. Every published limit shown is a nominal 5σ threshold at which recovery is not stated, which is what

this survey’s filled symbols are too; the open symbols above them are EIRP_{90} , the power recovered nine times in ten through the single-epoch trigger stage, and every limit in this paper is quoted on that. Any EIRP advantage toward the nearest systems is a d^2 effect and not an instrumental one. Panel (b) is the other axis: the channels searched here span 3.3 orders of magnitude, from 15.3 kHz to 31.25 MHz, and every one is far coarser than the 2.8 Hz channels of a dedicated narrowband backend. What distinguishes this survey is its frequency range and the proximity of its targets, not its depth in any one channel, and it complements the dedicated programmes rather than deepening them.

A non-detection becomes a quantitative statement only through a stated search volume. Wright et al. (2018) formalised that volume, and Margot et al. (2023) its conversion into an upper bound on the fraction of stars hosting a transmitter; Sheikh et al. (2025) asked where Earth’s own technosignatures would be detectable. Such a bound is valid only if the sample represents the population it is meant to describe. This sample was assembled by other observers for other purposes, so it does not, and §3 states what it is instead.

Because of the very specific and detailed statistical nature of this study, we have extended the paper to include several very detailed Appendices that rigorously explain the methodology involved.

3. THE SAMPLE

This search covers 1651 spectral windows toward 89 stars in 82 systems within 40 pc, drawn from 404 public ALMA execution blocks and 4.1 TB of raw data through one frozen pipeline, 280 h of telescope time and 1094 h summed over windows. We call that set the *primary census*: every limit, every crossing and every expectation here is computed over it, except that the crossing ledger also lists beside them the 2 crossings found in archival-tail blocks, which enter no expectation. Four further sets of blocks enter the paper without being part of it — a hold-out of 77 blocks and 315 windows, reserved before any block on its own side of the rule was searched; an archival extension of 115 blocks and 463 windows, searched afterwards; an out-of-sample control beyond 40 pc; and an external sample used only to calibrate the spatial statistic. Table 7 gives all five in rows that sum; wherever a number here is computed over something other than the primary census, that table’s name for it is used at the point of use.

Star means one stellar component, *system* a bound group counted once however many of its components were observed, and *target* one ALMA pointing.

The census is two experiments. 402 windows toward 65 stars in 60 systems are fine-channel — *Class A* throughout — and resolve a drifting carrier; every sensitivity and every exclusion quoted here refers to them unless stated otherwise. The remaining 1249 are coarse-channel, *Class B*, and support a secondary search for unresolved spectral excess that cannot distinguish a drifting signal from a stationary one.

The inclusion criteria were applied by query. A star enters if its Gaia DR3 parallax is at least 25 mas, measured to better than 20 per cent, and if its proper-motion-corrected position at the observation epoch falls inside the quoted field of view of at least one public execution

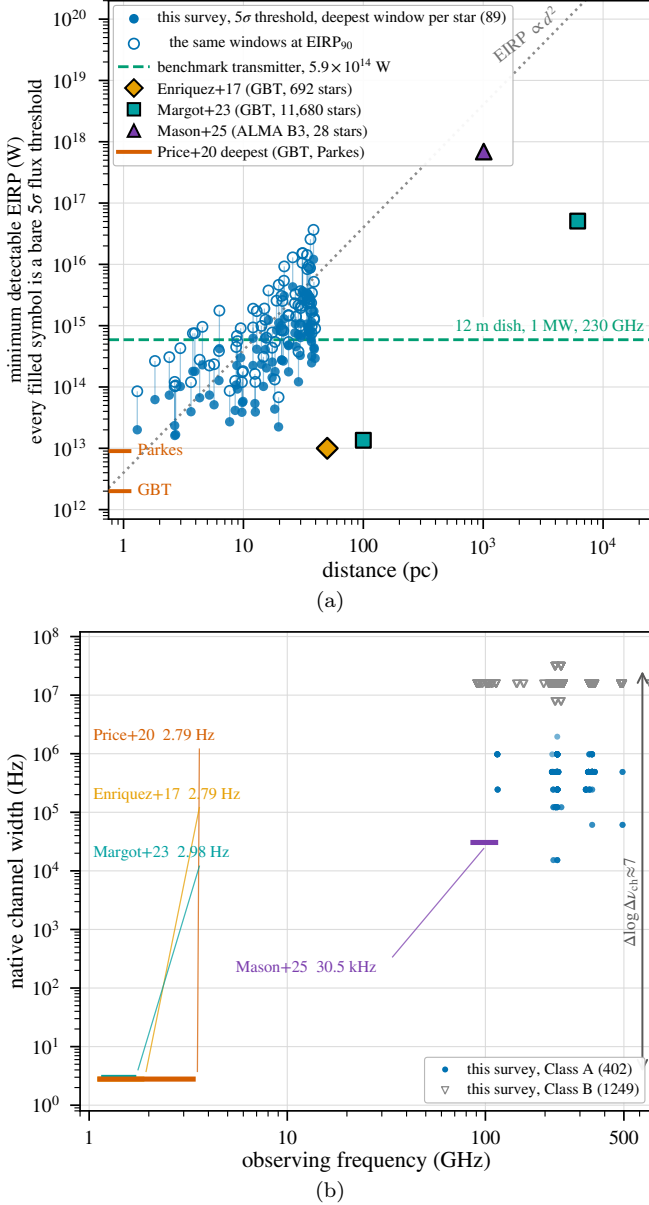


FIG. 1.— Where this survey sits among published radio technosignature searches. (a) Minimum detectable EIRP against distance. Blue, this survey, the deepest window per star: *filled*, that window's bare 5σ threshold, which is the quantity the other programmes quote; *open*, the same window at EIRP_{90} , the single-epoch trigger completeness of §5.1, joined to its own filled symbol. Orange, Enriquez et al. (2017); aqua, Margot et al. (2023); violet, Mason et al. (2025); left-edge ticks, Price et al. (2020), all at their own 5σ thresholds. Grey dotted, $\text{EIRP} \propto d^2$ at constant received flux; green dashed, the illustrative 12 m benchmark transmitter of §6.3. No symbol in the panel carries a confirmation criterion. (b) Native channel width against observing frequency for every searched window (filled, Class A; open, Class B), with the comparison programmes as horizontal segments. The $\Delta \log \Delta \nu_{\text{ch}} \approx 7$ annotation marks the span that opens once the comparison programmes' channels are set beside them: the channelisation, not the plotted power, is what differs most between programmes.

block. Distances are direct inversions, $d = 1/\varpi$, and the star need not sit at the pointing centre. One further rule applies to the window and not to the star: a window is kept only where the primary-beam response at the star is at least 0.5. It removes 8 windows, all of them ϵ Eridani in Band 6 at responses of 0.28–0.35,

and no others. The remaining 4 windows of that star and band place it at the pointing centre, at a response of 1.00: they pass the floor, and they were set aside with the rest of the band rather than by it, so Appendix A reports their limits and they enter no census statistic. The parallax criterion never binds: the worst fractional parallax error among the stars searched is 2.5 per cent, an order of magnitude inside it and inside the stricter $\sigma_\varpi/\varpi \leq 0.1$ of Table 1's reference census. The query returns 168 work-list entries; the quoted field of view is optimistic, and only 115 of those stars are in fact contained by a public field, of which 89 yielded a searchable window. One target carries no catalogue name and appears throughout under its ALMA source designation: ALMA J153702653–33192492 is Gaia DR3 6015967739554000768, an M5 dwarf at 35.0 pc.

Disc, planet and spectral type enter as descriptive metadata only, and no star was excluded by any criterion other than the three above. Three removals act on windows instead, and account for the whole difference between the 1721 rows the extraction wrote and the 1651 released: 45 repeated rows of one window, the 12 withheld above, and 13 windows failing a quality cut that is physical rather than chosen. Since $\sigma\sqrt{t_{\text{on}}\Delta\nu_{\text{ch}}}$ is set by system temperature and collecting area, a window more than 100 times below the sample median in it is not measuring the sky; the 13 removed lie 4.7×10^3 – 3.0×10^4 times below the median and the nearest retained 24 times, so any threshold between those removes the same windows. No integration-time cut was applied: the shortest window retained carries 21 s on source. The per-star and per-window ledger, including the whole decision tree and every edge case, is in the data release.

3.1. The archival selection

This sample was not designed: it is whatever previous ALMA programmes happened to observe, and 90 per cent of the stars searched were targeted under a debris-disc or planet-formation proposal. The bias is strong and measurable, and Table 1 tabulates it against a volume-limited reference census selected on the same parallax quality.

The dominant selection effect is distance. Stars within 5 pc are over-represented by a factor 49, and the enhancement falls monotonically with distance to a factor of two *under*-representation beyond 30 pc. Spectral type comes second: A stars are over-represented by about 23 and F and G stars by factors of two to three, because those are the stars with bright debris discs. M dwarfs are mildly depleted — 41 of the 89 searched stars are M dwarfs, a factor 0.8 of their share of the reference census, and only 29 of them carry the fine-channel coverage that can resolve a drifting carrier at all. Both ratios are lower limits, since Gaia temperatures are missing for about half the reference census and preferentially for its coolest members. Coverage is also concentrated: 285 of the 1651 windows come from 3 systems. Every statistical statement in this paper is therefore a statement about an archive-selected sample of nearby stars, and none of them is an occurrence rate for the solar neighbourhood. The selection acts hardest on exactly the targets a designed search would pick: of the 15 sample systems within 15 pc with a confirmed planet, listed in the data release, 6 carry no

TABLE 1

SELECTION FUNCTION OF THE SEARCHED SAMPLE AGAINST A VOLUME-LIMITED REFERENCE CENSUS, BOTH SELECTED ON THE SAME PARALLAX QUALITY: GAIA DR3 SOURCES WITH $\varpi > 0$, $\sigma_{\varpi}/\varpi \leq 0.1$ AND $1/\varpi \leq 40$ PC (17 566 OBJECTS, 8594 OF THEM CARRYING A `TEFF_GSPPHOT` ESTIMATE), AGAINST THE 89 STARS WITH AT LEAST ONE SEARCHED WINDOW. THE SEARCHED STARS ARE MEASURED FAR BETTER THAN THE CUT REQUIRES — THE WORST FRACTIONAL PARALLAX ERROR AMONG THEM IS 2.5 PER CENT — SO THE TWO COLUMNS ARE SELECTED ALIKE, AND LOOSENING THE CUT TO 0.2 ADDS NO CENSUS OBJECT AT ALL. 86 OF THE 89 SEARCHED STARS ARE CLASSIFIED: 53 FROM THE GAIA TEMPERATURE, 25 FROM THE SIMBAD SPECTRAL TYPE WHERE THAT TEMPERATURE IS ABSENT, AND 8 IDENTIFIED AS WHITE DWARFS; 3 CARRY NO CLASSIFICATION IN EITHER SOURCE.

THE SIMBAD ROUTE MATTERS BECAUSE `TEFF_GSPPHOT` IS MISSING PREFERENTIALLY FOR COOL STARS, WHICH IS THE POPULATION THE M DWARF STATEMENT IS ABOUT. “RATIO” IS THE SAMPLE COLUMN FRACTION OVER THE CENSUS COLUMN FRACTION; > 1 IS OVER-REPRESENTATION. *The strongest selection effect here is distance, not spectral type.*

Property	Census	Searched	Ratio
Distance			
$d \leq 5$ pc	48 (0.3%)	12 (13.5%)	49.3
5–10 pc	240 (1.4%)	13 (14.6%)	10.7
10–20 pc	2069 (11.8%)	18 (20.2%)	1.7
20–30 pc	5391 (30.7%)	19 (21.3%)	0.7
30–40 pc	9818 (55.9%)	27 (30.3%)	0.5
T_{eff} class ^a			
A	24 (0.3%)	5 (6.4%)	23.0
F	402 (4.7%)	9 (11.5%)	2.5
G	860 (10.0%)	16 (20.5%)	2.0
K	1400 (16.3%)	7 (9.0%)	0.6
M	5908 (68.7%)	41 (52.6%)	0.8
Other axes			
Confirmed planet hosts ^b	26 (15.5%)	23 (25.8%)	1.7
White dwarfs ^a	—	8 (9.0%)	—
Searched stars / systems	—	89 / 82	—

^aTemperature bins, not MK types: A ≥ 7500 , F 6000–7500, G 5300–6000, K 3900–5300, M < 3900 K, normalised over the 78 classified non-degenerate stars. The 8 white dwarfs are kept apart because a degenerate is not an early-type star and its photometric temperature would place it among them. Gaia `teff_gspphot` is unavailable for about half the reference census, preferentially for the coolest and faintest objects, so the census M fraction is a lower limit and every earlier-type ratio in this panel is correspondingly a lower limit; the searched column has no such gap.

^bBoth columns are the same positional join, of the 168-entry ALMA-covered work list and of its searched subset, to the NASA Exoplanet Archive `pscomppars` table accessed 2026 October 4; a confirmed planet is a row of that table. The 23 searched hosts carry 50 confirmed planets between them, and 21 of the 23 survive the deuterium-burning mass limit. The Gaia reference census carries no planet information, so the census column in this row is the work list rather than the 17 566-object Gaia census.

Class A coverage and so no drift-resolving limit.

3.2. Frequency coverage and archival scope

The Class A experiment covers $\Delta\nu_A = 47.7$ GHz of unique sky frequency, and that coverage is sparse: it is 26 disjoint islands scattered between 114.0 and 873.1 GHz, not a band: the endpoints are a range over which observations exist, never one that was covered. Adding the coarse-channel windows brings the total archival coverage to 118.1 GHz in 46 intervals, of which 113.1 GHz survives the molecular-line mask of §4. Fig. 2 shows the distribution; a transmitter at a frequency between the islands was never observable here, whatever its power.

The archive is not exhausted either, and the shortfall is accounted for. 177 further blocks lie inside the same query but were not processed when the census was frozen;

Table 7 gives what became of each. 144 have since been calibrated and searched, and 115 of those are reported here, as repeat epochs for the recurrence test and as a sample the frozen pipeline had not seen; the other 29 enter no number anywhere, for the reasons that table gives. The largest single gap is the 18 blocks for which the archive holds raw visibilities that were never pipeline-calibrated, 237 GB in all; no effort on our side would close it.

4. DATA AND METHODOLOGY

The search looks for one kind of signal: an unresolved carrier at the star’s own position, narrow compared with a single correlator channel, and drifting in sky frequency because the transmitter accelerates along the line of sight. Over a long track its power is spread across many channels, so a search that treats each channel on its own will miss it. The analysis de-drifts the data before it measures them.

4.1. The de-drifted search statistic

Each window’s dynamic spectrum is stacked along every trial drift rate $\dot{\nu}$. The search statistic is the largest stacked signal-to-noise ratio anywhere on the resulting channel $\times$ drift grid,

$$T(\mathbf{x}) = \max_{\dot{\nu}, \nu} \frac{\sum_t I(\mathbf{x}, t, \nu + \dot{\nu}t) / \sigma^2(t, \nu)}{[\sum_t 1 / \sigma^2(t, \nu)]^{1/2}}, \quad (1)$$

where $I(\mathbf{x}, t, \nu)$ is the spectrum at sky position $\mathbf{x}$ in integration t and channel ν , and $\sigma(t, \nu)$ is the noise scale there. Equation 1 is the matched filter for a constant-drift carrier: it adds the signal coherently along the track and divides by the noise of the stacked spectrum, so T is an absolute signal-to-noise ratio and not a local contrast. The noise scale is the median absolute deviation across 512 control positions at each integration and each channel, scaled to its Gaussian-equivalent standard deviation: local in time and frequency, global in position, and shared by the star and every control. The statistic is evaluated at one position at a time, identically at the star and at each control, which is what makes the two comparable. T_* denotes its value at the star.

Spectra are extracted at the star’s apparent position, never from images. For each integration and each native channel the pipeline forms the weighted vector average of the phase-shifted visibilities, in total intensity: the mean $I = (XX + YY)/2$ of the two parallel-hand correlations. That average is the dirty-map value at the extraction position — an amplitude, not a deconvolved flux density, so in a field with extended emission it is not a point-source measurement. The data are used as the archive delivered them, through ALMA’s own calibration and quality assurance, with their flagging carried through, every delivered channel searched, and no regridding, deconvolution or edge trimming. A per-integration median along the *frequency* axis removes the continuum; no median is taken along time, so a one- or two-channel feature cannot be removed by it. One subtlety matters for every power quoted below: the delivered channels are smoothed, so a channel’s effective noise bandwidth is not its separation and flux density times channel width is not yet a radiated power. A response correction computed

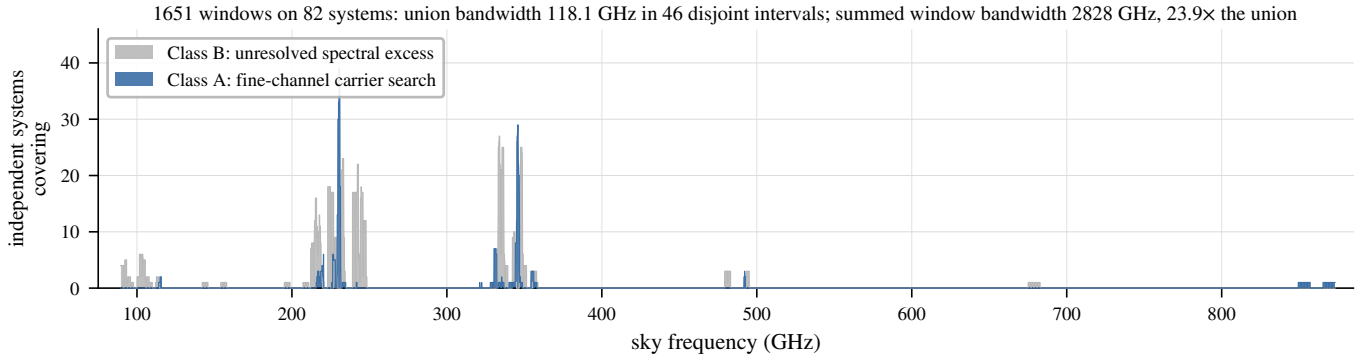


FIG. 2.— Where this survey searched. *Lower panel:* each searched window is a horizontal segment on its host system’s row, rows ordered by distance; heavy blue marks the fine-channel carrier search (Class A) and light grey the coarse-channel spectral-excess search (Class B). *Upper panel:* the number of systems covering each gigahertz of sky frequency; the threshold distribution is Fig. 1(a). The summed window bandwidth is 23.9× the union, so the union measures frequency coverage and the summed bandwidth search effort.

from the correlator’s own channel shape supplies the difference (Appendix A).

One convention holds everywhere in the paper. The search runs on the topocentric sky frequencies ALMA delivered, and every drift rate quoted is the apparent one at the telescope, which is what the data contain and what a reader can recompute from the released channel axes. Two steps need the star’s own frame and both take it: a crossing is compared with the line catalogue, and a repeat epoch is read at the predicted cell, after each block’s frequency axis has been corrected for the Earth’s motion at that epoch and then for the star’s systemic velocity.

Two power quantities carry the main text, both defined in Table 2: the trigger level $P_{\text{trig}} = 4\pi d^2 S \Delta\nu$ at distance d , for a carrier that makes $T_\star$ reach $T_{\text{trig}} = 5$ in one native channel of width $\Delta\nu$, and the sensitivity EIRP_{90} . The second is measured, not assumed: synthetic carriers of known amplitude were injected into calibrated visibilities and recovered through the unmodified pipeline. Three campaigns did that, each with its own denominator. The completeness itself is measured on 49 of the 402 fine-channel windows, injected directly, and on 5 936 carriers over 35 windows of the coarse class. What the spatial rank costs is measured separately, on 8 860 carriers over 52 of 56 attempted block–window configurations. The primary-beam correction is tested on a third set of 40 scored units, 10 of them also among the directly injected windows. Between them they span both ALMA arrays, four bands and the sample’s full range of channel widths. Every limit quoted in this paper is the EIRP_{90} of its own window, given to two significant figures beside its native channel width; the measured completeness is §5.1 and its error budget is Table 3. Two comparisons bear on the amplitude scale and they are not the same kind of check. An independent reduction of the same AU Mic visibilities is reproduced to 0.994 ± 0.014 , which tests the extraction and not ALMA’s flux transfer, an error in which would appear in both reductions alike; the external comparison is with the published quiescent flux density of ϵ Eridani, recovered to 1.10 ± 0.31 (Burton et al. 2022) and so a 28 per cent measurement. The absolute scale is therefore carried at ALMA’s own band-dependent accuracy specification, and the 14 Class A windows in Bands 8 and 10, at 492–493 and 849–873 GHz, accordingly carry ± 21 per cent where the sample as a whole carries ± 11 .

The drift range searched is bounded by an acceleration,

not by a frequency. Drift rate and line-of-sight acceleration are one quantity in two sets of units,

$$\frac{\dot{\nu}}{\nu} = \frac{a_{\text{los}}}{c} \quad \text{and so} \quad a_{\text{los}} = \frac{c\dot{\nu}}{\nu}, \quad (2)$$

so a ceiling expressed as $\dot{\nu}/\nu$ is the same physical limit in every band. The drift and acceleration ceilings released with the catalogue were computed from Equation 2 window by window, and reproduce it to 1×10^{-4} over all 402 Class A windows. The default ceiling is $12.0\text{--}13.3 \text{ Hz s}^{-1} \text{ GHz}^{-1}$, the acceleration of a transmitter orbiting a solar-type star at a few hundredths of an au. It is the Mason et al. (2025) fiducial with a threefold margin, widened for a confirmed exoplanet host whenever the innermost known planet demands more; at unit margin the search would have been blind to two of the windows that produced a crossing. It is far narrower than the general recommendation of Sheikh et al. (2019), which is set by solar-system rotation rather than by a planet’s own acceleration, and the restriction is deliberate. The grid is uniform in $\dot{\nu}$, and fine enough that a carrier midway between two trials loses no more than 0.26 of a native channel over the longest track. Two limits follow: a deliberately chirped carrier is not distinguished from an accelerating platform, and a transmitter need not be bound to a known planet, so drifts faster than the ceiling are unconstrained (Fig. 3).

Only part of the archive can carry a drift search at all, and the two classes reported here are that division (Table 2). Channelisation in the sample is bimodal with an empty gap between the modes, so the split is unambiguous, and it coincides with the physical criterion that matters: whether the maximum searched drift crosses of order one channel or more over the on-source track, $\eta_{\text{drift}} = |\dot{\nu}_{\text{max}}| \tau_{\text{track}} / \Delta\nu_{\text{ch}} \gtrsim 1$. Each window’s own η_{drift} , trial-drift count and acceleration ceiling are released catalogue columns, so a reader may reclassify on that criterion instead; the two do not partition the survey identically. The two classes are physically different experiments, results are given per class, and they are combined only where that is appropriate.

4.2. From a crossing to a confirmed signal

Every window passes through the same three tests in the same order, and three words name what comes out of them: a *crossing*, an *unattributed crossing* and, had one survived, a *confirmed signal* (Fig. 7). Each is set

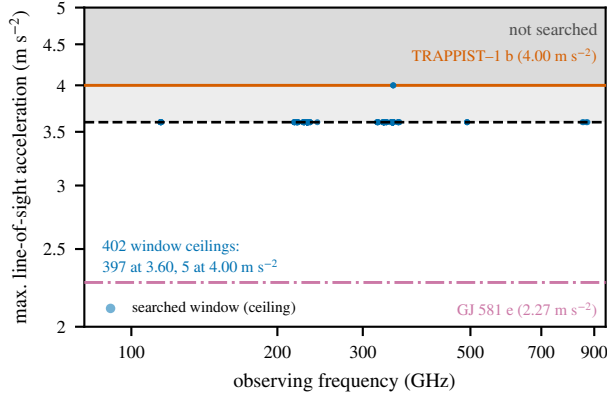


FIG. 3.— What the drift search reaches, as a physical selection function. The drift ceiling of each of the 402 Class A windows (points), converted to line-of-sight acceleration by Equation 2, against observing frequency; the shaded region above is unsearched. The boundary is flat because the grid was built at constant acceleration; in the units the search works in it is not, running 1.4 kHz s^{-1} at 115 GHz to 10.5 kHz s^{-1} at 872 GHz . Horizontal lines are reference accelerations: Earth’s rotation at the equator (0.034 m s^{-2}) and orbit (0.0059 m s^{-2}), and the two most demanding known planets of any target, TRAPPIST-1 b (4.00 m s^{-2}) and GJ 581 e (2.27 m s^{-2}). The boundary drawn is the default ceiling, 3.60 m s^{-2} , which is also the median; it runs to 4.00 m s^{-2} on the 5 TRAPPIST-1 windows. For the most demanding case, transiting TRAPPIST-1 b, 29 per cent of the orbital phase lies above the default ceiling and 1 per cent above the widened one, which falls 0.01 per cent short of that planet’s own maximum acceleration.

out below as the quantity it measures, the number that quantity is compared with, when that number was fixed, and what passing it does and does not establish. There is no fourth test, and none of the three is a significance test.

- (i) *Trigger* $\rightarrow$ *a crossing*. The quantity is T_* , the de-drifted statistic of Eq. 1 read at the stellar position; the criterion is $T_* \geq T_{\text{trig}} = 5$ in at least one channel $\times$ drift cell; and both the statistic and the level were fixed before the archival search began. T_{trig} is a **trigger threshold and not a detection significance**: T is not shown to follow any standard distribution under the null, so the level carries no tail probability, and a window holds between 112 and 6.87×10^6 of those cells, so one level buys very different protection in a coarse single-pointing window and in a finely channelised one on a long track. A crossing establishes only that the window goes on to the next two tests.
- (ii) *Line attribution* $\rightarrow$ *an unattributed crossing*. The quantity is the crossing’s velocity offset from the nearest transition of a molecular-line catalogue, evaluated in the star’s own rest frame; the criterion is $\pm 50 \text{ km s}^{-1}$. The statistic and the trigger were fixed before the archival search began; the line catalogue is a single database query made afterwards, on 2026 October 6, so the dispositions rest on the physical evidence set out beside each one and not on the order in which the list was assembled. The half-width follows the Keplerian line-of-sight velocities of gas surviving beyond a few au, with margin for the systemic velocity. A crossing inside that window is *attributed* where the species is one that such systems are known to carry: a coincidence with a catalogued transition makes an astrophysical origin far more likely than a

transmitter, the more so where a second transition of the same species or known circumstellar emission is present. A further three crossings coincide with a species no debris disc has been shown to carry; their offsets are reported and they are counted as unattributed (Appendix C). A coincidence is in any case a statement about a catalogue rather than a property of a frequency, so this is an attribution and not an exclusion, and **every crossing goes on to the recurrence test whichever side of the window it falls**, wherever a later block covers the frequency. An unattributed crossing is a target for follow-up and not a detection. The window’s price is that the survey constrains nothing inside it (Appendix C).

- (iii) *Independent recurrence* $\rightarrow$ *a confirmed signal*. The quantity is T_* at the cell the first epoch predicts, read in another execution block of the same star at a single drift rate after the frame correction of §4.1; the criterion is that it reach the trigger again; and the test was specified before the repeat epochs were searched. It is a weak criterion, and the archive’s uneven temporal sampling is why: only part of the sample has epochs separated well enough to apply it at all.

Two diagnostics hang off the second test, and the word is exact: each says which crossing deserves attention first, and neither decides anything. The first ranks T_* against the same statistic at 512 control positions in the same window, drawn once from a fixed seed into an annulus placed by each block’s own synthesised and primary beams so that no control sits on the star; its lowest value, $1/(512 + 1)$, means the star beat every control. It is a ranking filter and not a significance test, because the stellar and control positions are not demonstrably exchangeable: out of sample, on 315 windows the frozen pipeline had never seen, the stellar rank has median 0.405 where one half is expected if the two were interchangeable, and a parameter-free correction for the annulus geometry does not restore an exchangeable statistic (Appendix B). It also responds to compactness rather than to origin, as β Pictoris shows: genuine carbon monoxide ranks below its own control ring there, because the resolved disc lifts the ring along with the star. The second diagnostic is a visibility-domain fit, which asks whether a crossing’s emission is at the stellar coordinates rather than elsewhere in the field; it cannot discriminate at the trigger either, because a noise maximum at the star has flux at the star (§5.5). **Neither gates a disposition, so neither is charged to the sensitivity**, and the limits here are correspondingly deeper than a chain charging for the rank would give. Throughout, p denotes a genuine significance level, and no probability computed against the rank’s own null is called one.

One execution block is not a null, and a uniform rule says so. The comparison with the control positions needs each block’s ensemble to behave like noise, and a block fails when the median control statistic over its own windows, scaled by the median of that quantity over its class, exceeds twice that scale. Of the 404 primary-census blocks, 1 fails — A002_Xc7fa6f_X28a8 toward BD+05 1668, at 4.1 where the next worst is 1.12. The Class A results are identical with it and without it, the

same 402 windows and 52 crossings either way, because its 4 windows are all coarse-channel; what it does hold is 4 of the 56 crossings, the survey’s entire Class B crossing population (Appendix D).

Because a fixed threshold is not a fixed false-alarm rate, the number of crossings expected by chance is computed window by window, from the exceedance rate measured at each window’s own control positions; the reserved blocks then test the one assumption that makes, which is that a star behaves like a control position in its own field (§5.3.1).

Three corrections precede the statistic: the extraction field is chosen so that the star lies inside the delivered primary-beam field and not merely inside the quoted field of view; the stellar position is propagated to the observation epoch for proper motion, though not for the annual parallax, whose omission is corrected window by window in §5.1; and on-source time is accounted per integration from the delivered exposure records rather than from the scan intentions. That the chain finds what it is meant to find is shown twice over, by the injected carriers above and by β Pictoris CO — a real feature at a known stellar position, recovered and localised in two transitions and 9 execution blocks (§5.4).

5. RESULTS

The primary experiment is the Class A search for a drifting, unresolved spectral carrier: 402 fine-channel windows toward 60 systems. The Class B search for an unresolved spectral excess is secondary and carries no drift discrimination: 1249 coarse windows. Together they are the 1651 windows of the released catalogue. **No event satisfied the confirmation criterion set out in §4.2.** What follows is first how deep the search reaches, and then what it found and how each finding was disposed of.

5.1. Sensitivity: the power at which a carrier would have been found

The sensitivity of this survey is a measured completeness of one epoch, and not a threshold. Artificial carriers were injected at a random sub-channel phase and a random drift from each window’s own grid and recovered through the unchanged pipeline; the power at which nine in ten come back is EIRP_{90} , the only sensitivity quoted here. *What it measures is recovery through the single-epoch trigger stage* — the trigger and the localisation at the stellar position, inside one execution block. **It is therefore the power at which a carrier would have produced a crossing and not the power at which one would have been confirmed**, because confirmation needs a second and separated epoch, which the archive supplies for some systems and not others whatever their depth (§5.6.1). The 512-position spatial rank decides nothing and is not charged against it either: charging it would cost a further factor 1.31 and in 3 bright-ring windows could not be met at any injected power (Fig. 4).

Measured that way the completeness is one number. Nine carriers in ten are recovered at $3.03 P_{\text{trig}}$, and that class-average factor is determined to ± 2 per cent. Every one of the 49 directly injected Class A windows resolves a ninety per cent point of its own, and the value does not depend on the brightness of its control ring: the 5

TABLE 2

THE QUANTITIES AND THE THREE WORDS THIS PAPER USES, DEFINED ONCE. EVERY TERM BELOW IS USED IN THIS SENSE THROUGHOUT AND IN NO OTHER. THE DISTINCTION THE FIRST THREE ROWS DRAW IS THE ONE THE LIMITS OF §5.1 TURN ON: A CROSSING IS A SINGLE-EPOCH EVENT, AND A CONFIRMED SIGNAL NEEDS A SECOND AND SEPARATED EPOCH THAT NO SENSITIVITY SUPPLIES.

Term	Definition
Crossing	A channel $\times$ drift cell at the stellar position reaching $T_{\star} \geq 5$ in one execution block, and specifically the largest such cell in that window: the search records one statistic per window, so a window whose maximum is a molecular line can hold no weaker carrier at all. An event requiring investigation; no claimed significance.
Unattributed crossing	A crossing lying more than 50 km s^{-1} in the star’s frame from every transition of the line catalogue. That is a reproducible frequency coincidence; which coincidences are astrophysically plausible is a separate judgement, stated with each one. A target for follow-up; not a detection.
Confirmed signal	A crossing recovered at the predicted stellar-frame frequency and drift in an independently separated block. The only confirmation criterion, applied to every crossing: one coincident with a catalogued transition would meet it too and be reported as a recurring line. It can be applied only where the archive observed the same frequency again, so for the 21 Class A systems where it did not, no sensitivity makes a confirmed signal possible. Over the $\pm 833 \text{ km s}^{-1}$ also searched about that cell, a pair of crossings is tested against a single bounded trajectory where the drift grid resolves the interval between them (7 of 13 pairs, Table 4); over longer baselines no pair can be separated and the comparison is a population one, against the number chance predicts.
P_{trig}	The power of a carrier that makes $T_{\star}$ reach the trigger in one native channel. An internal pipeline level: not a sensitivity, not a significance.
EIRP_{90}	The power at which nine injected carriers in ten produce a crossing at the star: a measured single-epoch trigger completeness, not a detection probability. Measured for the 49 injected Class A windows; for the other 353 an estimate from the class-average calibration.
Class A	The primary experiment: 402 fine-channel windows, median 488.3 kHz, toward 60 systems over 47.7 GHz of union bandwidth, in which the searched drift crosses of order a channel or more.
Class B	The secondary experiment: 1249 coarse-channel windows, mode 15.625 MHz, constraining a persistent unresolved excess. No drift discrimination, and never a carrier search.
The sample	Neither volume-limited nor representative of the solar neighbourhood: the stars ALMA happened to observe for other reasons (§3).

windows whose ring carries resolved emission come back at $3.24 P_{\text{trig}}$ against 2.94 for the rest (Fig. 4). Class B carries its own factor $4.17 P_{\text{trig}}$, measured the same way from 5936 carriers injected into 35 coarse windows of 14 execution blocks; it is what sets the limits for Barnard’s Star, Wolf 359 and Proxima.

The resulting limits. The median Class A window reaches 2.9×10^{15} W of equivalent isotropic radiated power and the median system, on its best window, 1.6×10^{15} W, with a spread across systems of 6.8×10^{13} – 3.7×10^{16} W (Fig. 5a). The round 10^{15} W level is reached for 25 of them. The calibration of the factor is good to $\times 0.89$ – $\times 1.11$ (Table 3), and that is the interval which applies to the median and to the counts. One bias is not inside it and runs one way only: an injected carrier is added to already calibrated visibilities and so suffers no atmospheric decorrelation, whereas a real carrier does. That loss is measured block by block from the phase rms the observatory records on its phase calibrator, a point source at the phase centre which loses coherence exactly as a carrier would (Cortes et al. 2026, §10.4.9), through the standard relation between phase scatter and visibility amplitude (Richards et al. 2022, Eq. 3); it leaves the median limit optimistic by 1 per cent and the median window above 400 GHz by 8 per cent, so the limits at the top of the frequency range are the least secure quoted here.

Carriers were injected directly into 49 of the 402 Class A windows, 12 per cent, and the result transferred to the remaining 353, which is the dominant caveat on all of the above. The injected windows are the Class A windows of execution blocks drawn in a stratified random order fixed in advance, without reference to which windows hold a crossing; they hold 7 crossings against the 6.3 the population rate predicts. *The unit is the block and not the window*, because a block’s calibrated visibilities are what an injection campaign has to rebuild: the draw controlled band and array, while channel width, ring brightness and integration length vary inside a block and matched the population afterwards to 22 per cent on the worst axis.

A window that was not injected therefore carries an estimated EIRP_{90} , and across the injected windows the ninety per cent point ranges from $\times 0.80$ to $\times 1.58$ of the adopted factor: that dispersion, not the calibration interval above, is what one uninjected window’s own limit is uncertain by. Because a system’s limit is its best window, a minimum over several such estimates could favour windows whose own factor falls below the class average; drawing each window’s factor from the 49 measured ones instead, over 4000 trials, moves the per-system median deeper by 2.5 per cent, keeps 25 systems at 10^{15} W (24–25 in nine trials in ten) and spreads one system’s own limit by -17 to $+9$ per cent. The estimate is mildly conservative and the statistics formed over the class are unaffected.

One geometric term acts window by window, and it is measured rather than modelled. The phase centre assumed in the search omits the annual parallax, which displaces the star by a median 0.022 of a synthesised beam and at most 0.67; each limit is divided by the amplitude a compact source retains at its own displacement (Appendix A), which makes the median Class A limit shallower by 1.5 per cent. One window, Wolf 28

TABLE 3

WHAT SETS EIRP_{90} , AND IN WHICH DIRECTION. THE UNCERTAINTY ON THE CALIBRATION IS AN UNCERTAINTY ON THE CLASS-AVERAGE FACTOR, AND SO ON ANY STATISTIC FORMED OVER MANY WINDOWS; THE FLUX TERM WITHIN IT IS BAND-DEPENDENT, SO THE SUBMILLIMETRE WINDOWS CARRY THE LARGER OF THE TWO COMBINED FIGURES. THE SPREAD BETWEEN WINDOWS IS AN OBSERVED DISPERSION AND IS THE STATEMENT ABOUT ONE WINDOW’S OWN LIMIT; THE SIGNED TERMS ARE BIASES, STATED WITH THEIR DIRECTION RATHER THAN FOLDED INTO AN ERROR BAR. THE DECORRELATION BIAS IS MEASURED IN 279 OF THE 404 EXECUTION BLOCKS, THOSE WHOSE DELIVERY RECORDS THE PHASE STABILITY, AND IS NOT IMPUTED TO THE REST.

	value	what it does to the limit
Class-average completeness	$\times 3.03$	nine carriers in ten recovered at this multiple of a window’s own trigger power, over 49 injected Class A windows
<i>Uncertainty on that calibration (per cent), combined in quadrature</i>		
The factor itself	± 2	sampling error of the pooled value over the 49 injected windows
Absolute flux scale	± 10.2	the observatory’s own band-dependent specification (ALMA Proposer’s Guide): ± 5 in Bands 1–5; ± 10 in Bands 6 and 7; ± 20 in Bands 8–10, weighted over this sample’s bands
Distance and visibility scale	± 4.5	the Gaia parallaxes, which enter as d^2 , and the block-to-block visibility scale
Combined	$-11 / +11$	$\times 0.89$ – $\times 1.11$ on the factor, and so on the median limit and on the system counts; ± 21 for the 14 Class A windows in Bands 8 and 10, at 492–493 and 849–873 GHz
<i>Spread between windows. An observed dispersion, not an error bar.</i>		
Window to window	$\times 0.80$ – $\times 1.58$	the 49 injected windows resolve 2.41 – $4.80 P_{\text{trig}}$ individually, so an uninjected window’s own completeness is not known to the interval above
<i>Signed terms. A bias is not an error bar and is not added to one.</i>		
Channel response	$\times 1.93$	carried by the injections, so already inside the completeness above
its residual	$+8$	the injected profile falls short of the exact response; conservative
Primary-beam model	$+0 / +10$	ALMA’s beamwidth in place of the pipeline’s Airy-null one: nothing in the median window, 4 windows of one star
Decorrelation	$+0 / +1$	measured block by block from the phase rms the observatory records on each block’s phase calibrator, which an injected carrier does not suffer and a real one does: $+8$ above 400 GHz, and more than $+20$ in 5 per cent of windows

in A002_X9f2ff8_X25b0, has no astrometric keys and is left uncorrected.

A carrier that is not always radiating is penalised in proportion. The de-drifted stack is coherent across a whole execution block, so a carrier present for a fraction of one comes back at that fraction of its continuous amplitude. Over 3888 trials at duty cycles down to 10 per cent the recovered amplitude followed the duty cycle to 1 per cent. A transmitter radiating a tenth of the time therefore needs $10\times$ the power quoted above, and because those trials carry no drift the joint drifting and intermittent case is not measured here at all.

5.2. The nearest stars: Barnard’s Star and Wolf 359

Barnard’s Star is the second-nearest stellar system and Wolf 359 the third if brown dwarfs are excluded, and to our knowledge these are the first millimetre or submillimetre technosignature limits for either (Enriquez et al. 2017; Price et al. 2020; Margot et al. 2023). Both are non-detections. *Both limits come from the secondary Class B experiment and not from the carrier search:* the archive

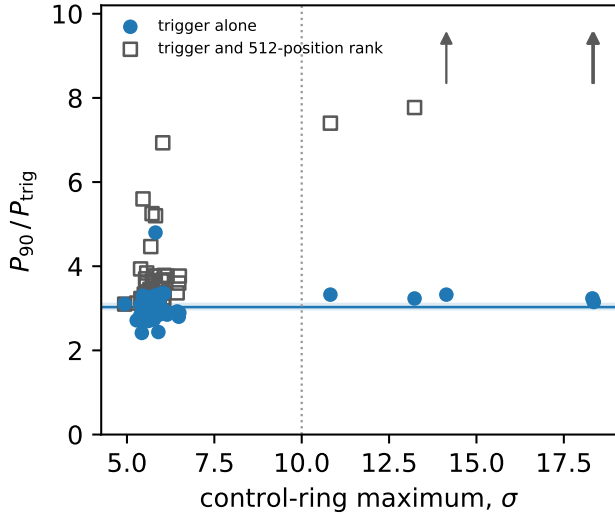


FIG. 4.— Why one completeness factor serves the whole class. Each of the 49 of the 402 Class A windows that were injected directly is plotted at the power that recovers nine carriers in ten, against the maximum of its own 512-position control ring. Filled circles are the criterion adopted here — the trigger and localisation at the star — with the pooled value and its uncertainty as the line and band; they show no trend with the ring. Open squares add the demand that the carrier outrank every control position, and arrows mark the windows that then reach ninety per cent at no injected power. Dotted, the 10σ ring brightness above which a ring carries resolved emission rather than noise.

holds only coarse Band 6 windows toward these stars, so what is constrained is an unresolved spectral excess and not a drifting carrier. On the EIRP₉₀ convention of §5.1 Barnard’s Star reaches 2.7×10^{14} – 4.0×10^{14} W over 12 windows and Wolf 359 3.1×10^{14} – 4.5×10^{14} W over 12.

5.3. Threshold crossings and how each was disposed of

The result is a chain of three tests, and it is read in one direction. Of the windows the search covers — the 1651 of the released catalogue and the archival-tail blocks beside them — 56 reach $T \geq 5$ at the stellar position. The search records one statistic per window, the largest value its per-channel drift-maximised statistic reaches at the star, and a *threshold crossing* is that maximum where it reaches the trigger: at most one to a window, with no weaker cell beside it. 4 lie in the one execution block that fails the quality criterion of §5.3.2, fixed before it was applied, so the population characterising the search is the 52 crossings in quality-passing data and those 4 are reported beside it rather than within it. The molecular-line mask, $\pm 50 \text{ km s}^{-1}$ wide and evaluated in each star’s own rest frame, places 21 of the 52 within that distance of a catalogued transition; three of those coincidences are with a species no debris disc has been shown to carry and are reported rather than acted on (Appendix C), leaving 18 attributed and 34 unattributed, every crossing in one column or the other. Of the unattributed, 2 outrank all 512 control positions of their own window, and none was found again in any later epoch covering its frequency. **No crossing survives the chain, and this survey reports no detection.** Figure 7 carries the chain with the number surviving each step and Tables 9 and 10 the crossing-by-crossing ledger behind it.

The attributions are kinematic and not an accident of bandwidth, but a window whose maximum is a line

costs the whole window. Of the 51 crossings whose measured peak frequency the release records, 11 fall within $\pm 50 \text{ km s}^{-1}$ of a searched transition against 4.7 expected were each peak free to land anywhere on its own window’s channel grid (Appendix E). A weaker carrier in a window whose maximum is a line could not have been recorded at all, so the 21 such windows were searched again with the masked channels removed; all but 3 could be, and the re-search leaves one further cell at the trigger, matched by the control positions of the same windows and absent from the blocks that cover it again (Appendix C).

5.3.1. Most of the crossings are expected by chance

The trigger is a threshold on a statistic and not a false-alarm rate: the number of searched channel $\times$ drift cells per window spans 4.8 decades, so one level buys very unequal protection. How many crossings chance should produce is therefore measured window by window, over the 402 Class A windows of the census and with one condition on both sides of the comparison: a cell at the stellar position that reaches the trigger and that the line mask does not attribute. A window’s own 512 control positions measure how often a position in that field reaches the trigger, and the window’s own channel grid measures how often such a position would fall inside an attribution window, which takes 20 per cent of the total. Chance predicts 28.9 unattributed Class A crossings, with 12–48 allowed by resampling the execution blocks the windows come from, and 32 are found. A crossing is a window’s own maximum (§5.3), so each side of the comparison counts windows and the two count the same thing. A further two unattributed crossings, both toward HD 14055, lie in archival-tail blocks the catalogue does not tabulate window by window, and enter neither side.

The expectation is a sum of rates measured at control positions, and the reserved blocks measure how far a star departs from them: carrying that difference through moves it to 30.8, and the count agrees with chance either way. The width of the range allowed is what the comparison can say and no more — an added population of as many as 19 crossings would sit inside it unremarked. What tests each crossing individually is the recurrence test of §5.6.

5.3.2. One execution block accounts for four of the crossings

4 crossings belong to BD+05 1668 (GJ 273, Luyten’s Star), all of them in one ACA execution block, A002_Xc7fa6f_X28a8, whose whole field is above threshold: 512 of 512 control positions exceed $T = 5$ against a survey median control maximum of 2.75. The single-integration noise is as predicted (0.1609 Jy against 0.1611 Jy) but the residual does not integrate down — a jackknife on the channel mean gives 0.030 Jy against 0.0048 Jy thermal — so a slow drift common to every channel and position inflates the statistic across the field, and the 24 sibling executions of the same field, array and spectral setup reach only $T_\star = 4.26$ and produce no crossing. *What is inflated is the field and not the star.* The strongest of the four reaches $T_\star = 73.4$ where its own control ensemble peaks at 74.7, and between 1 and 421 of the 512 control positions stand above the star in these windows, so there is nothing here to localise and no evidence either way about a carrier. Applied block by block the same criterion fails this block and no

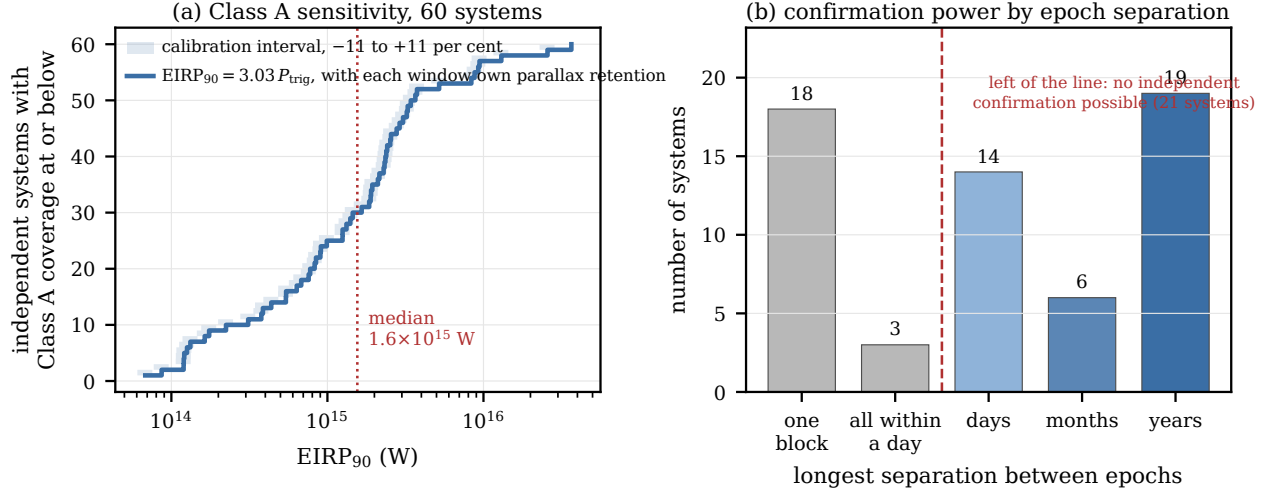


FIG. 5.— **The two completenesses of the carrier search, over the same 60 systems.** (a) Single-epoch trigger completeness: the cumulative number of systems whose best window reaches a given EIRP₉₀, the power at which nine carriers in ten produce a crossing at the star inside one execution block, on the one adopted convention $3.03 P_{\text{trig}}$ with each window's own annual-parallax retention applied; the shaded band is the calibration interval of Table 3. (b) Confirmation completeness, which no amount of power buys: systems grouped by the longest separation between their searched epochs, more than 1 day counting as independent. **For the 18 observed in a single block and the 3 whose blocks all fall within one day — left of the dashed line, 21 systems — confirmation was not possible at any sensitivity in panel (a):** what panel (a) gives them is a crossing limit, not a detection limit.

other (Appendix D.2), which is why its 4 crossings are reported apart from the 52 in quality-passing data, and Appendix E takes what the block might instead be to the windows in which the star lies well off the phase centre.

5.3.3. Stellar flares are excluded by a broadband prediction

Stellar activity is not excluded by the sample, which contains known flare stars. But a millimetre flare raises every channel of a window alike, so one able to produce a single-channel crossing at the trigger must produce a $5\sqrt{N}\sigma$ continuum event in the same integrations. Over the 48 crossings with a recoverable continuum measurement that prediction runs 49–4,717 σ , against an observed continuum statistic of median -0.29 with $|Z| < 3$ in 45 of them. One block settles it: AU Mic A002_Xa42f75_X319 holds both a crossing ($T_\star = 5.27$) and the published millimetre flare of MacGregor et al. (2020), and *that real, bright, independently published flare contributes 0.17 σ to the single-channel statistic* — raising the continuum at $Z = 10.3$ while the crossing's own de-drifted track reaches $Z_{\text{max}} = 2.1$.

5.4. β Pictoris: an astrophysical positive control, and the obstacle it identifies

A search that finds nothing must show that it could have found something, and the survey's strongest features show exactly that. The frozen pipeline recovers the β Pictoris circumstellar carbon monoxide blind, without being told the disc is there. It recovers it in two rotational transitions and in 10 independent execution blocks spanning 8.4 yr, and in 9 of those the star also outranks every control: 5 Band 3 windows at $T_\star$ up to 30.76, and 4 Band 6 windows at 11.68. The identification is kinematic and does not rest on those statistics. Carried through the frame chain — to the barycentre at each block's own epoch, then to the star's rest frame — all 10 crossings lie within 0.5 km s⁻¹ of the stellar systemic velocity, in the two lowest CO transitions of a star with mapped circumstellar CO (Dent et al. 2014; Matrà et al. 2017). The

Band 3 peaks agree across blocks to 1.00 channel once each block's own tuning is removed, so any apparent drift is far below the grid searched. The alternative reading requires a transmitter placed by coincidence at matched offsets from two CO transitions of one star.

It is also the control that shows the recurrence test of §5.6 working in the positive direction: every flagged β Pictoris carbon-monoxide feature is recovered in an independent execution block, the screened Band 6 window returning $T_\star = 8.95$ in the second block of its own recurrence set. A non-recurrence elsewhere means nothing unless the same machinery recovers a repeat where one exists.

One of those blocks is a counterexample, and it is the more useful half of the control. In it several of the 512 control positions exceed the stellar statistic, so the spatial rank disfavors a window holding carbon monoxide known to be real and centred on the star; the disc is resolved on the scale of the primary beam in that configuration, which is the geometry the statistic is built to reject. The rank is therefore a test of compactness and not of authenticity, and on a real extended source it errs conservatively — a stronger statement than a clean positive control, because it bounds the diagnostic's reach as well as demonstrating it (Fig. 10).

HD 48370 is the complementary case and is equally instructive. Its crossing reaches $T_\star = 26.83$, but the control ring rises with the star, to 18.36, as emission filling the primary beam does. The velocity matches the prominent CO(2→1) that Cataldi et al. (2023) report toward the source and flag as possible foreground cloud emission. The block's ¹³CO(2→1) window is bright at the ring and faint at the star. What identifies the extended emission is therefore the *level* of the control ring and not the star's rank against it — the star here outranks every control — which is why the window-level flag of Appendix D.2 is defined on the ring's median.

Taken together, these two results identify the dominant obstacle for millimetre technosignature searches, and it

is not sensitivity. The strongest apparent signals in this survey are circumstellar and foreground molecular emission, and 18 of the 56 crossings land on a catalogued transition. The mask that removes them costs 4.3 per cent of the unique-frequency union over 12 species, and it is nevertheless the single most productive test in the chain. The warning for future work is about line lists rather than about depth, and Appendix C puts a number on it.

5.5. *The two crossings that lead their own control fields*

Two crossings reach a stellar statistic above every one of the 512 control positions in their own window: 61 Vir at 346.313811 GHz and CP-72 2713 at 344.269746 GHz, both in Band 7. They are drawn in Fig. 6 and tabulated in Table 5, and neither is a detection. The control rank is recorded for them because it says which crossing was examined first; it disposes of nothing (§4). CP-72 2713's crossing lies -12 km s^{-1} from SO(8₈-7₇) in the star's own frame, a coincidence reported as a note rather than as a disposition (Appendix C). Both events go on to the recurrence test.

Neither returns when the same stellar-frame frequency is observed again. 3 other execution blocks cover 61 Vir's cell. A carrier persisting at the discovery flux would have given $T_{\text{pers}} = 9.7$ in their combination, which reads $T_{\text{rep}} = +0.39$ at the carried drift, so the two readings of one amplitude disagree by 4.9σ on Eq. B1; searching the whole grid at that cell instead of the one carried drift turns up nothing above 2.71, which is a maximum over every trial and is not a quantity T_{pers} may be set beside. CP-72 2713 is covered by 5 later blocks, the deepest beginning 2.05 hours afterwards and 1.1 times deeper than the discovery; combined they give $T_{\text{pers}} = 6.3$ against -0.82 , or 4.8σ . Appendix B follows that crossing through every step of the chain.

The visibility-domain re-measurement sits beside the chain rather than inside it. 51 of the 56 crossings were re-fitted by phase rotating to the star and following the recovered drift, and 41 are consistent with a point source there; but a noise maximum *at* the stellar position has flux at the stellar position, so at the trigger the fit returns a real part close to the trigger itself, and injected sources in the bin these crossings occupy return $\text{Re}/\sigma = 6.58 \pm 1.97$. Both events are consistent with a real source and neither excludes one; **the discriminating evidence is the line mask and recurrence.** (The experiment uses three control sets of three sizes: 512 image-plane positions in the rank screen, 8 of them with a retained dynamic spectrum, 12 reachable in the visibilities.)

5.6. *Recurrence, applied to every unattributed crossing*

A carrier that stays inside the repeat window and radiates through both observations must be there when the same frequency is observed again. That is the confirmation criterion, applied in the star's own frame: the predicted channel follows from the Earth's motion at the repeat epoch and the star's systemic velocity, the discovery drift is carried to it, and the statistic is read there, maximised over nothing. The frame matters: for η Crv it moves the predicted channel by -80.2 channels, so a sky-frequency test would have read a channel holding no information about the event.

One cell combined over many blocks is a narrower model than persistence. The cell is 488 kHz wide, 0.4 km s^{-1} , so combining the reading over every covering block by inverse variance requires the carrier to return to that same stellar-frame cell at the start of each of them while drifting by kilohertz a second within each — a periodic reset, not simply a transmitter that stays on. The combination is therefore quoted beside the nearest single repeat block, which assumes only that the carrier was there at both epochs: for HD 14055, whose exclusion rests on 30 blocks spanning 653 days, the nearest block alone rules out a carrier above 0.91 of the discovery amplitude and the combination above 0.36 of it. Across the 37 crossings with repeat coverage the medians are 0.95 and 0.38 of that amplitude, and for 15 the nearest block rules out nothing.

One cell also assumes more than the data support, because a drifting carrier is a carrier that is accelerating. The drift grid reaches 3.6 m s^{-2} of line-of-sight acceleration, and the largest drift fitted to any of these crossings lies at that ceiling, which is therefore where the bound belongs. A body held there by a star of $2 M_{\odot}$ or less circles it every 3.6 days at 176 km s^{-1} , so its carrier need not come back to the same cell at all. What is searched at each repeat epoch is therefore the whole spectral window at every drift in the grid, $\pm 833 \text{ km s}^{-1}$ about the predicted cell, which spans that excursion for 30 of the 32 crossings a window of another block also covers. Those windows number 328, of which 300 hold nothing above the trigger anywhere at any drift.

The 28 that do hold something are themselves crossings, and that alone does not dispose of them: a carrier present again would appear in its repeat block as a crossing, so to set one aside for being a crossing is to assume the answer. Their number agrees with the 28 chance predicts in trials of that size, but that is a statement about the set. Each pair must be asked on its own whether one transmitter could have produced both members, and Table 4 asks it of all 13 pairs, across 5 stars, whose two crossings lie within 833 km s^{-1} of each other in one star's frame. The two fitted drifts and the elapsed time fix the offset a carrier at constant line-of-sight acceleration would show, leaving nothing free, and the comparison resolves the interval for 7 of the 13 pairs. All 7 fail, by 3.2 standard deviations in the mildest case. *The test is not one that can only say no:* the HD 285968 crossings that are present again, 5.0 days apart in CO(2-1) at one stellar-frame frequency to 2 kHz, are the one set in this survey it accepts, at 0.9σ .

β Pictoris is the closest the data come to a pair, and it is excluded three ways. Its two crossings sit at 241.7739 and 241.8594 GHz in blocks 24.4 hours apart, 85 MHz or 106 km s^{-1} apart, so one trajectory would have to climb at a mean $+972 \text{ Hz s}^{-1}$. Both fitted drifts are negative, -1625 and -692 Hz s^{-1} : the carrier was falling in frequency at both epochs and would have had to rise between them, so the pair fails on the sign before any arithmetic. Constant acceleration then places the second crossing -102 MHz from the first where it is observed $+85$, a miss of 187 MHz or 232 km s^{-1} ; absorbing that would need both drift fits wrong by 2131 Hz s^{-1} , enough to carry the carrier across 47 channels of its own dwell and leave neither crossing to be found. Letting the acceleration vary does not rescue it either: the gentlest trajec-

tory joining the two drifts and the observed offset needs its line-of-sight acceleration to cross the whole searched range in 16.4 hours, against the 21 hours a quarter of the fastest permitted orbit takes.

Over longer baselines the comparison is a population test and not a per-pair one. For the other 10 pairs, 2 days to nearly two years apart, a bounded trajectory does exist: the orbital freedom the acceleration ceiling allows over those intervals is wider than the pairs are separated, so nothing we can measure tells one carrier from two. The closest such case, 57 km s^{-1} apart in blocks 1.9 days apart, misses constant acceleration by 292 MHz but needs its acceleration to turn over in 38 hours, which an orbit can do. What disposes of those pairs is the agreement of their number with chance, and Table 2 and Fig. 7 say so.

No crossing was found again. All 38 unattributed crossings were tested, over 444 readings in 151 execution blocks. At the predicted cell the largest statistic over all of them is $+2.00$ at the carried drift and 4.71 over every drift trial, against a trigger of 5. Of the 38, 37 retain a discovery amplitude and so can carry an exclusion, and 4 of those are withheld because that amplitude comes from the block failing the quality criterion of §5.3.2, which leaves 33 quoted exclusions. Over those 33 the repeat coverage rules out a carrier at a median 0.43 of the discovery amplitude and the deepest at 0.017 of it, but the individual limits run to 1.00, and for 3 of them — β Pic at 241.75 and 241.84 GHz and HD 110411 at 219.45 GHz — nothing below 0.84 is ruled out. As two measurements of one amplitude, discovery and repeat disagree by $3.4\text{--}5.6\sigma$, median 4.8σ : the most the pair can say when the discovery amplitude is itself a five-sigma measurement. Where the criterion has little power its silence is not an exclusion, and the chance a persisting carrier would have satisfied it follows from the $T_\star$ and T_{pers} columns of Table 9: above 90 per cent for 29 of the 33, and 0.41 for β Pic at 241.75 and 241.84 GHz, ALMA J153702653–33192492 at 347.22 GHz and CD–57 1054 at 346.51 GHz, which are untested rather than excluded.

The last crossing, HD 14055 at 331.1559 GHz, retains no discovery spectrum and so carries no exclusion; the criterion itself needs only a frequency and a drift grid, and 15 of the 17 windows covering it hold nothing above the trigger at all.

5.6.1. What can be searched, and what can be confirmed

A sensitivity limit and a confirmation limit are different statements, and this experiment has two extents. It searched 60 systems and 402 Class A windows over 47.7 GHz of union bandwidth; it could have confirmed a signal in 37 of those systems, the ones observed again at the same frequency and more than a day apart, and in 11 of them more than a year apart (Fig. 8). The difference is the cost of retuning.

An independent execution block is not an independent epoch. Every test above uses a block other than the discovery block, but the blocks are not all separate in time: the baselines run from 0.46 hours to 1.8 yr, and for 21 of the 37 crossings carrying an amplitude the nearest covering block falls within a day. Over the hours to days where there is repeat coverage these tests constrain a carrier that stays in the searched window and radiates

TABLE 4

CAN TWO CROSSINGS BE ONE CARRIER? EVERY PAIR OF CROSSINGS TOWARD ONE STAR, IN TWO EXECUTION BLOCKS, WITHIN 833 km s^{-1} OF EACH OTHER IN THE STAR'S FRAME; STARS ARE IDENTIFIED BY POSITION, SO THE TWO ARCHIVAL-TAIL CROSSINGS OF HD 14055 PAIR WITH ITS CATALOGUED ONES. $\Delta\nu$ IS THE STELLAR-FRAME OFFSET AND $\dot{\nu}_1, \dot{\nu}_2$ THE DRIFTS FITTED INDEPENDENTLY IN THE TWO BLOCKS. A TRANSMITTER AT CONSTANT LINE-OF-SIGHT ACCELERATION WOULD PUT THE PAIR AT $\Delta\nu = \frac{1}{2}(\dot{\nu}_1 + \dot{\nu}_2)\Delta t$; THE LAST COLUMN GIVES HOW FAR THE PAIR IS FROM THAT, IN UNITS OF THE DRIFT GRID'S OWN RESOLUTION CARRIED OVER Δt . ITALIC ENTRIES ARE PAIRS FOR WHICH THAT RESOLUTION EXCEEDS THE WINDOW SEARCHED, SO THE COMPARISON HAS NO POWER AND ONLY THE AGREEMENT OF THE PAIRS' NUMBER WITH CHANCE BEARS ON THEM. THE LAST ROW IS THE CONTROL: THE ONLY CROSSINGS IN THIS SURVEY THAT ARE PRESENT AGAIN, WHICH THE SAME TEST ACCEPTS.

Star	blocks	Δt (d)	$\dot{\nu}_1$ (Hz s^{-1})	$\dot{\nu}_2$ (Hz s^{-1})	$\Delta\nu$ miss (MHz) (σ)
61 Vir	X822/X82f	3.00	+1052	+3182	+755.4 16
ALMA J1537–3319	X10359/X398	123	–2023	+2373	+223.5 3.2
AU Mic	X1838/X319	310	+1419	+2627	–159.0 28
HD 14055	X2b2ee/X2c3bb	0.09	–2426	+4184	+825.2 2448
	X24be0/Xd46	1.92	+1104	–3857	+63.0 42
	X2c3bb/X3835	2.00	+4184	+1769	–514.6 143
	X2b2ee/X3835	2.09	–2426	+1769	+310.6 49
	X3835/Xd526	607	+1769	–2589	–555.5 8.9
	X2b2ee/Xd526	609	–2426	–2589	–244.9 56
	X3835/X25c5e	649	+1769	+2584	+423.3 49
	X2c3bb/X25c5e	651	+4184	+2584	–91.3 76
	X2b2ee/X25c5e	651	–2426	+2584	+733.8 1.5
β Pic	X3a90/Xa9df	1.02	–1625	–692	+85.5 132
HD 285968	X108f/X5b4e	4.99	–49	–48	–0.002 0.9

through both observations; over months and years they bound no duty cycle; and for the 36 systems with no second block covering a frequency the first one covered they bound nothing at all.

5.6.2. The reserved hold-out and the control beyond 40 pc

Two sets of execution blocks were searched with the frozen pipeline and then reserved for calibration (Table 7); both hold data on real targets and both are reported here as searches. The hold-out — 77 blocks, 315 windows, 36 stars and 214 h on source, withheld by a rule fixed before the search finished and sharing no execution block with the census (Appendix B.1) — contains 8 threshold crossings, the strongest at $T_\star = 8.64$, and *not one outranks its own control positions*: the fewest controls above the star in any of them is 5 of 512. 7 were tested for recurrence on the same code path as the census, and exactly one is present again: HD 285968 at $+9 \text{ km s}^{-1}$ from CO(2–1), whose 4 later blocks combine to $T_{\text{rep}} = +6.71$ at the carried drift, the value Table 6 prints, while the strongest single block reaches 5.87 at its own best drift. It is a molecular line, and *that* it recurs is the test working in the positive direction; for every other tested crossing the repeat coverage rules out a carrier at 0.47 of the discovery amplitude or less (Table 6).

The control beyond 40 pc is 47 blocks and 198 windows toward 22 background sources, 88 h on source, with 8 crossings whose strongest is $T_\star = 12.7$ and none of which outranks its own controls. A systemic velocity is known for only 0 of them, so it supports no limit and shows only that the pipeline behaves the same way on fields it was not tuned for. The third reserved set calibrates the spatial rank and is not external at all (Appendix B).

5.7. A stellar-frame multi-epoch stacked search

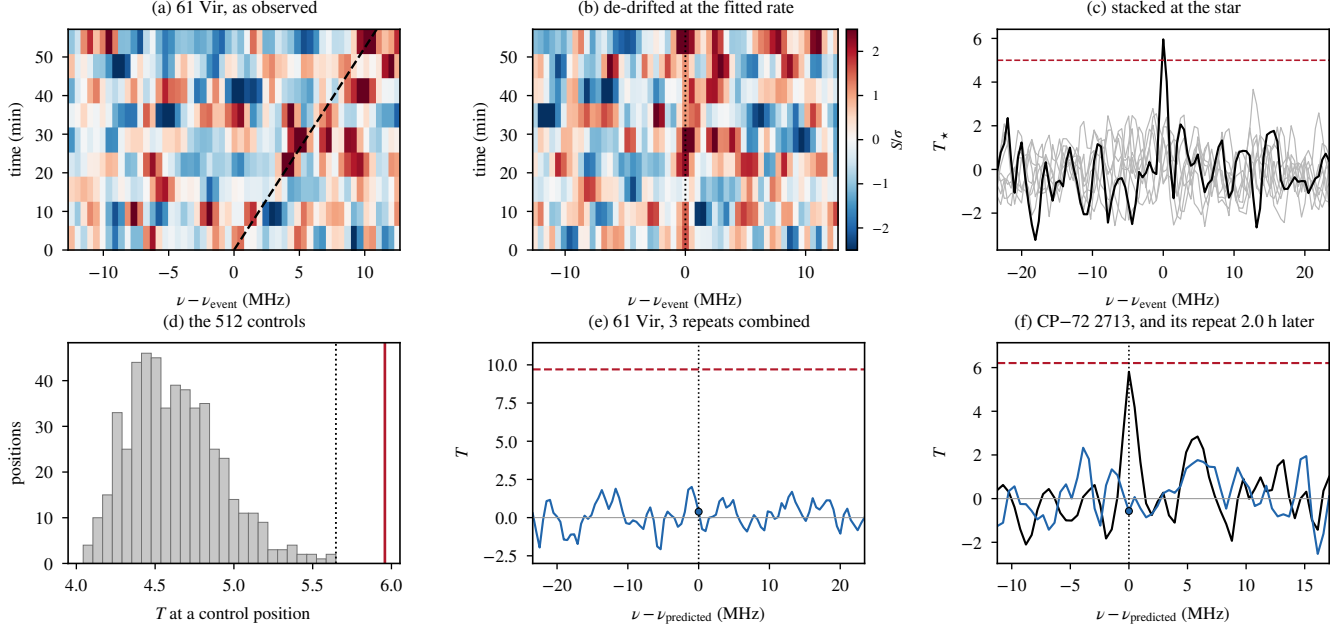


FIG. 6.— **Why this survey reports no confirmed signal although it produced an interesting event.** (a) The dynamic spectrum at 61 Vir’s position around the crossing at 346.313811 GHz, observed 2017 May 20, with the fitted drift of $+3182 \text{ Hz s}^{-1}$ as the dashed track; the 225 integrations are binned into 9 and plotted in units of each bin’s noise. At $119 \pm 20 \text{ mJy}$ against 303 mJy per integration the carrier is 0.39 standard deviations in any one of them: it exists only in the coherent stack of all 38 minutes on source. (b) The same rows shifted by that drift, which carries the carrier 22 channels of 488.3 kHz across the 57 minutes the observation spans; 8 of the 9 bins are positive at the marked channel. (c) The de-drifted stack at the star (black) and at the 8 control positions with retained dynamic spectra (grey), in units of its own noise; dashed, the trigger. (d) The same statistic at each of the 512 rank-screen control positions: the star (solid) against the highest control (dotted). (e) The 3 other blocks covering that cell in the star’s rest frame, combined, with zero at the predicted channel, which the frame transport moves between epochs (Table 5 dates them). Dashed, the $T_{\text{pers}} = 9.7$ a carrier persisting at the discovery flux would have given; circled, the $T_{\text{rep}} = +0.39$ read there at the carried drift, 4.9σ away on Eq. B1. (f) The second event, CP–72 2713 at 344.269746 GHz: its discovery stack (black) and the same cell 2.05 hours later in a block 1.1 times deeper (blue), with the same two marks at $T_{\text{pers}} = 6.2$ and, in that one block, $T_{\text{rep}} = -0.57$ — not the combination over all 5 covering blocks that Table 5 prints. The two events imply isotropic-equivalent powers of $5.1 \times 10^{14} \text{ W}$ at 8.5 pc and $8.9 \times 10^{14} \text{ W}$ at 36.7 pc .

Combining the epochs of a star in that star’s own rest frame deepens the limit. 55 of the 82 surveyed systems hold more than one searched execution block, and 58 of the surveyed stars can be stacked, 5 of them from blocks beyond the census. Over 315 stacked groups, 2177 epoch-windows and 1426 h, the flux limit improves by a median factor 1.52 over each star’s best single epoch and 4.45 at most. 11 further stars stack too — 10 control stars beyond 40 pc and 1 covered only by archival blocks — and are excluded from those figures; over all 69 stacked stars, whose stacks are mostly shallow two-epoch ones, the median gain is 1.46. Every stacked limit is measured by injection into the stack itself and never read off a stacked spectrum as a threshold.

A fixed stellar-frame frequency is not a fixed observed frequency, so the epochs are registered before they are added. The barycentric term alone swings by a median 8.2 km s^{-1} between the epochs of one star and by as much as 53.9 — many channels at the 488 kHz Class A width — so epochs added as observed would dilute rather than reinforce. Each epoch is transformed to the stellar rest frame, divided by its own primary-beam response and combined by inverse variance over the channels of common stellar-frame coverage, keyed on the star’s catalogued identity rather than the directory the search ran in. Spectra with pathological archived visibility weights are excluded first (27 of 2838, which would otherwise dominate the sum). The transform is verified on a real line: HD 285968’s molecular line collapses from a

19.26-channel topocentric spread to 0.798 channels over 4 epochs spanning 44 d.

The criterion was fixed before any stellar position was examined: a stacked $Z \geq 6.0$ above every stacked control maximum, no epoch supplying more than 0.60 of the weighted sum, and at least 0.50 of the signal surviving removal of the strongest epoch. Controls are stacked identically and every clause evaluated at them, measuring the whole test’s false-alarm rate at 0.21 per cent against a 1 per cent ceiling fixed in advance.

The gain is $\sqrt{N_{\text{eff}}}$ and not $\sqrt{N}$, where $N_{\text{eff}} = \sigma_{\text{min}}^2 \sum_e \sigma_e^{-2}$ is the equivalent number of equal-depth epochs and the median N_{eff}/N is 0.65. What a stack realises is $\sqrt{N_{\text{eff}}}$ divided by the measured noise of its own stacked spectrum, and that noise follows the inverse-variance prediction to 0.992 in the median, with a per-stack scatter of 7 per cent and nine stacks in ten between 0.81 and 1.18. Against $\sqrt{N}$ the realised gain is 0.78, the shortfall being unequal epoch depth, so an epoch count alone misleads. HD161868 stacks 13 epochs for a gain of only 1.15: its N_{eff} is 1.7, so $\sqrt{N_{\text{eff}}}$ is 1.29, and its stacked spectrum came out 1.12 times as noisy as predicted.

The deepest stacked limit is Proxima Centauri’s, and it is a Class B limit. Proxima is covered by 66 blocks over 85 h spanning 6.3 yr, all of them coarse 15.625 MHz windows, so the stack constrains an unresolved spectral excess and discriminates no drift. 25 of those blocks belong to the primary census, 33 were searched afterwards from the same public archive and 8 come from the

TABLE 5

THE TWO EVENTS OF FIG. 6. THE CROSSINGS WHOSE STELLAR STATISTIC EXCEEDS ALL 512 CONTROL POSITIONS OF THEIR OWN WINDOW. REPEAT EPOCHS MEASURED ARE THE BLOCKS AT WHICH THE PREDICTED STELLAR-FRAME CELL WAS READ; THE INDENTED ROWS GIVE HOW MANY OF THOSE THE RELEASED CATALOGUE CARRIES AND HOW MANY BLOCKS IT LISTS AS COVERING THE FREQUENCY AT ALL, WHICH DIFFER BECAUSE THE ARCHIVAL EXTENSION IS DEPOSITED SEPARATELY. T_{rep} COMBINES EVERY REPEAT EPOCH BY INVERSE VARIANCE AT THE CARRIED DRIFT: IT IS ONE CELL AND ONE TRIAL, SO IT MAY BE NEGATIVE, AND IT IS THE QUANTITY T_{pers} IS COMPARED AGAINST. THE INDENTED READINGS BENEATH IT ARE THE LARGEST SINGLE EPOCH AT THAT DRIFT AND THE LARGEST OVER EVERY DRIFT TRIAL, EACH A MAXIMUM AND SO POSITIVE ON AVERAGE WHATEVER IS THERE; THE SINGLE-EPOCH READING IN THE DEEPEST BLOCK IS THE ONE FIG. 6(F) MARKS, AND IT IS NOT THE COMBINATION. THE EXCLUSION IS EQ. B1, AND BENEATH IT THE FRACTION OF THE DISCOVERY AMPLITUDE S STILL RULED OUT AT THE TRIGGER, $(T_{\text{rep}} + 5)/T_{\text{pers}}$, AND THE CHANCE $\Phi(T_{\text{pers}} - 5)$ THAT A CARRIER PERSISTING AT S , AND THEN AT THE TRIGGER AMPLITUDE $5S/T_{\star}$, WOULD HAVE SATISFIED THE CRITERION AT ALL.

	61 Vir	CP-72 2713
Frequency (GHz)	346.313811	344.269746
Channel width (kHz)	488.28	488.28
Drift rate (Hz s ⁻¹)	+3182	-2056
$T_{\star}$	5.958	5.809
Highest control T	5.65	5.68
Flux density (mJy)	118.8 ± 19.9	11.3 ± 2.0
Peak-channel EIRP (W)	5.1×10^{14}	8.9×10^{14}
Repeat epochs measured	3	5
in the released catalogue	3	1
catalogue blocks at ν	4	1
Observed	2017 May 20	2022 Oct 2
Nearest covering epoch	2017 May 19	2022 Oct 2
separation (h)	-1.97	+2.05
covering epochs more than a day away	2	4
Longest baseline (d)	3	2
T expected if persistent, T_{pers}	9.70	6.30
T measured, matched drift, T_{rep}	+0.39	-0.82
largest single epoch	+1.65	+0.96
maximised over every drift	2.71	3.60
Exclusion, σ_{excl}	4.9 σ	4.8 σ
Amplitude still excluded at 5σ	0.56 S	0.66 S
Recovery chance if persistent, at $S / \text{at } 5\sigma$	1.00 / 1.00	0.90 / 0.66
Disposition	not confirmed	not confirmed

reserved hold-out, all observed over the census blocks' own years. Stacking the census blocks alone, the set every other limit here uses, gives $\text{EIRP}_{90} = 2.2 \times 10^{13}$ W; adding the rest of the archive deepens that by 1.26 to 1.73×10^{13} W, against 5.51×10^{13} W for the best single epoch on the same injection-calibrated convention. Its N_{eff} is only 7.9, so $\sqrt{N_{\text{eff}}} = 2.80$ of the measured 3.19 is what combining the epochs buys, the rest being this spectrum's own noise. The limit holds for an emitter steady in the stellar frame to within one 15.625 MHz channel, 20.9 km s^{-1} peak to peak, with no inter-epoch drift above $7.8 \times 10^{-2} \text{ Hz s}^{-1}$. At 1.30 pc that depth is mostly geometry; the gain is what stacking bought.

2 groups pass the full criterion and both are known lines. They are β Pictoris' CO $J = 1-0$ at $+3.09 \text{ km s}^{-1}$ over 8 epochs ($Z = 45.3$) and its $J = 2-1$ at $+3.55 \text{ km s}^{-1}$ ($Z = 12.1$), agreeing to 0.46 km s^{-1} or 0.73 of a CO(1-0) channel: the astrophysical positive control passing in the stack. The survey's own $\pm 50 \text{ km s}^{-1}$ mask leaves 0 of them unattributed. HD 285968 reaches $Z = 12.89$ on the line used above as the registration standard and is rejected by its own controls at 14.19: the emission is extended, as a cloud is and a transmitter is not.

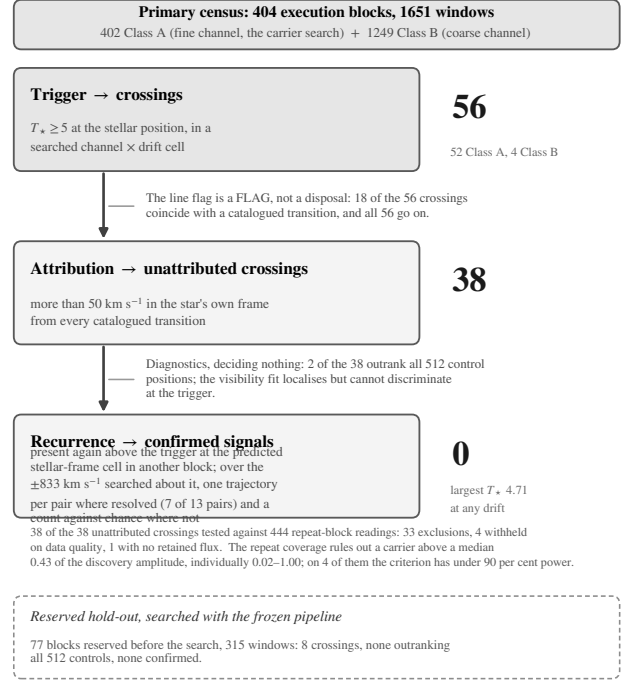


FIG. 7.— The detection logic, with the count at each level. Three levels and no more: a *crossing* is an event requiring investigation, an *unattributed crossing* is a target for follow-up, and a *confirmed signal* would be a detection. Coincidence with a catalogued transition is a flag and not a disposal, so all 56 crossings continue past it; the spatial rank and the visibility fit are diagnostics attached to the middle level and decide nothing. The reserved hold-out is drawn as a separate branch because a transmitter in it would matter whatever its calibration role. Every count is read from the ledger, which lists every crossing the search produced, including those in the archival-tail blocks and the quality-failing block that no expectation is computed over, so the figure, Table 9 and the text cannot disagree.

The restriction travels with every number above, and its width is the channel. At CO(2-1) one 488 kHz channel spans 0.63 km s^{-1} , the whole excursion a stacked carrier is allowed, peak to peak. A transmitter on a short-period giant planet, its line-of-sight velocity swinging by $\pm 30 \text{ km s}^{-1}$, crosses 94 of those channels peak to peak and de-registers completely, against 3.0 of a 15.625 MHz window: the coarse stacks buy reflex tolerance with bandwidth. Drift divides the same way, a stack in 488 kHz channels being confined to $1.3 \times 10^{-1} \text{ Hz s}^{-1}$ over the median inter-epoch span of 44 d. No crossing of $\$5.6$ drifts that slowly: the slowest measured is 519 Hz s^{-1} , which leaves one channel in 16 minutes, and in 33 of 38 cases the stack covering a crossing contains its discovery block. Toward the 20 stars carrying them the stack is a deeper limit on a non-drifting carrier, not a test of these events. Each stacked limit therefore constrains only emitters stable in the stellar frame to within its own channel; the single-epoch limits of $\$6.3$ apply generally.

6. DISCUSSION

6.1. What is different about a millimetre technosignature search

The most transferable result of this experiment is not the non-detection but a measurement of what limits a carrier search at these wavelengths. For classifying a crossing in the present archival data set, molecular-line confusion and sparse repeat coverage are more immedi-

TABLE 6

THE THRESHOLD CROSSINGS OF THE RESERVED HOLD-OUT, SEARCHED WITH THE SAME FROZEN PIPELINE AND DISPOSED OF BY THE SAME TESTS AS THE CENSUS. Δv_* IS THE OFFSET FROM THE NEAREST MASKED TRANSITION IN THE STAR'S OWN FRAME AND n_{rep} THE NUMBER OF COVERING REPEAT BLOCKS; T_{pers} AND T_{rep} ARE AS IN TABLE 9. THE ONE CROSSING THAT IS PRESENT AGAIN IS A CARBON MONOXIDE LINE.

Star	block	ν_{topo} (GHz)	T_* line	Δv_*	n_{rep}	T_{pers}	T_{rep}	disposition
HD 285968	Xdaef5a Xdd9b	230.5081	8.642 CO(2–1)	+8.7	4	11.6	+6.71	attributed; present again
HD 14055	X107376c X6126	330.7123	5.513 ^{13}CO (3–2)	+112.6	30	312.0	+0.56	unattributed; does not recur
TRAPPIST-1	Xd0e450 X259b	346.2454	5.326 SO(9 ₈ –8 ₇)	-306.6	5	11.1	-0.32	unattributed; does not recur
HD 172555	X107fa96 X1300f	344.6066	5.144 SO(8 ₈ –7 ₇)	+251.3	—	—	—	unattributed; no repeat coverage
HD 285968	Xdb4b9a Xa7c	229.9129	5.106 CO(2–1)	-768.0	4	11.2	+0.21	unattributed; does not recur
HR 1010	Xc6b674 X23f2	346.2301	5.044 SO(9 ₈ –8 ₇)	-238.7	7	25.9	-0.80	unattributed; does not recur
HD 10647	Xff99e1 X9cdf	345.7059	5.017 SO(2 ₃ –2 ₁)	+37.1	13	142.1	-0.93	unattributed; does not recur
HD 170773	Xff294f X1fb	345.5597	5.001 SO(2 ₃ –2 ₁)	-113.8	4	59.3	+1.45	unattributed; does not recur

ate limitations than thermal sensitivity. That is a statement about disposing of events and not about depth: the power a transmitter must radiate to be found here remains a median 1.6×10^{15} W per system.

Confusion, and not interference, sets the false-positive rate. Catalogued molecular transitions, almost all of them carbon monoxide, account for 18 of the 56 threshold crossings, and no crossing in this survey is attributed to interference (Appendix E). 10 of the 12 windows that reached the spatial stage are disc or foreground carbon monoxide, arising in 2 of the 69 debris-disc systems searched. The remedy that works is a velocity-space mask evaluated in the stellar frame, and it is not free: it removes 4.3 per cent of otherwise usable coverage.

Spatially extended emission is the second discriminant, and it has no analogue at centimetre wavelengths. A line filling the primary beam raises the star and the control annulus together instead of cancelling, so the spatial statistic loses power exactly where circumstellar gas is brightest. β Pictoris shows the benign face: the frozen pipeline recovers its circumstellar carbon monoxide blind, in two transitions, which is the positive control the experiment needed. HD 48370 shows the other. Its resolved disc emission lies near a band edge, and that one window contributes 59 per cent of every near-edge threshold exceedance in the survey: pooled, the near-edge rate is $\times 2.464$ the interior rate; without that window $\times 1.020$, and on a held-out half $\times 1.181$ ($p = 0.24$). What looked like a property of band edges is one disc, so no edge correction is applied to any limit.

The channelisation is inherited. 1249 of the 1651 windows have channels so coarse that every drift in the grid stays inside one channel over a whole track; they measure an amplitude but cannot distinguish a drifting carrier from a stationary one, which is why the carrier search rests on 402 of them. The widths on offer here run from 15.3 kHz to 31.25 MHz, about 3.3 orders of magnitude; set beside the 2.8 Hz channels of centimetre-band carrier searches the range becomes 7 orders (Fig. 1(b)), and nothing in the search could choose them.

Repeat epochs turn a crossing into a testable claim, and an archive supplies them only incidentally. Let the confirmation coverage $C_{\text{conf}}(\nu)$ be the number of systems toward which two independent execution blocks more than a day apart both cover the sky frequency ν , so that a carrier found once there could have been sought again. It must be stated against frequency rather than as a fraction of the sample, because repeat visits re-tune and a second epoch is of no use at a frequency it does not cover. Its support is 24.7 GHz of union bandwidth, 52 per cent of the 47.7 GHz searched, falling to 8.5 GHz beyond a

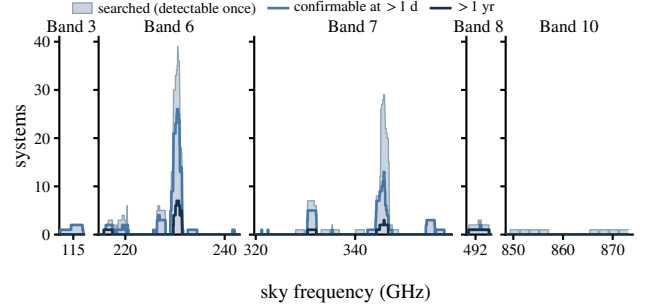


FIG. 8.— What could be searched against what could be confirmed. For the Class A experiment, against sky frequency: the number of systems toward which a carrier could have been detected once (grey), the number for which a second independent execution block more than a day later also covers that frequency (blue), and the same beyond a year (dark). One panel per populated ALMA band, widths in proportion to the frequency each contributes, as in Fig. 9.

year, and at its best near 231 GHz it reaches 26 systems (Fig. 8). The searched extent is about twice the extent over which a persistent carrier could have been confirmed, and for close to half the sample the archive constrains an intermittent emitter hardly at all, whatever its power.

Finally, a threshold is not a sensitivity. A trigger level is a property of the noise, not of what a transmitter of a given power does to a pipeline: measured by injection and recovery through the unchanged selection, the power needed for ninety per cent recovery is 3.03 times a Class A window's own trigger power and 4.17 times a Class B window's.

6.2. The search space this archive opens

The carrier search is 402 fine-channel windows toward 60 stellar systems, 47.7 GHz of union bandwidth; the 1249 coarse-channel windows are a supplementary spectral-excess experiment. No previous search has looked for technosignatures toward stars within 40 pc above about 30 GHz, and the one earlier millimetre search worked in Band 3 toward far more distant stars (§2); Band 3 is searched here, in 112 windows down to 90 GHz including β Pictoris' CO $J = 1-0$ control. Of the 10 MHz–115 GHz haystack axis of Wright et al. (2018) this experiment covers 19.9 GHz, only 1.0 GHz of it carrier search because the Class A union begins at 114 GHz; the rest lies above that axis.

Union bandwidth overstates what that buys: different frequencies were searched toward different numbers of stars and to very different limits. Figure 9 gives instead the number of independent systems that would have yielded a recovery at ninety per cent complete-

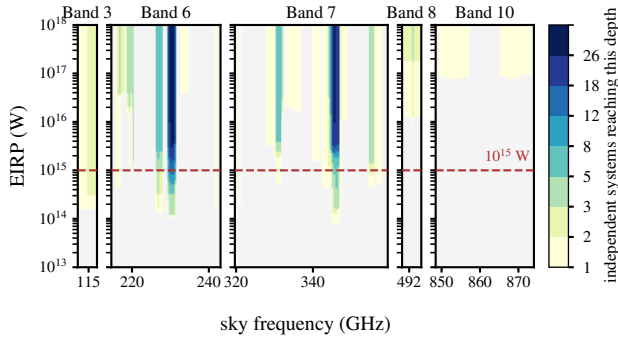


FIG. 9.— Where the experiment could have found something. Shading gives the number of independent systems that would have yielded a recovery at 90 per cent completeness for an unresolved carrier of the plotted EIRP at the plotted sky frequency; unshaded means none. One panel per populated ALMA band, with panel widths in proportion to the frequency each contributes: the 47.7 GHz Class A union is 26 disjoint islands across 114.0–873.1 GHz, which a continuous abscissa would draw as hairlines. The dashed line is 1×10^{15} W. Figure 2 shows where ALMA looked; this shows where the experiment was sensitive.

ness, against sky frequency and transmitter power. At an EIRP of 1×10^{15} W the covered frequency falls to 19.1 GHz, 40 per cent of the union, reaching 25 systems and 17 at the best-covered frequency near 231 GHz; with no limit on power that frequency reaches 39 of the 60 systems, and half the sample only above 3.4×10^{15} W.

Archival SETI buys sky and frequency cheaply and buys confirmation not at all: 82 systems and 118.1 GHz of union bandwidth across both classes without an allocation, on data selected by others and so carrying the biases of §3.1.

The conventional measure for comparing programmes is the continuous-waveform transmitter figure of merit of Enriquez et al. (2017), $\text{CWTFM} = \zeta_{\text{AO}} \text{EIRP}_{\text{min}} / (N_{\star} \nu_{\text{rel}})$ with $\nu_{\text{rel}} = \Delta\nu / \nu_{\text{mid}}$, smaller being better. Those programmes quote EIRP_{min} at a bare detection threshold, so the Class A experiment is put on that footing: with $N_{\star} = 60$ and $\nu_{\text{rel}} = 0.097$ it is 1.0×10^5 on the sample’s largest threshold power and 4.4×10^3 on the median system, against 0.85 for that survey. Margot et al. (2023) publish no CWTFM; the same expression on the three quantities they do give — 11 680 stars, 1.15–1.73 GHz and $\text{EIRP} > 1 \times 10^{13}$ W — returns 0.11. With the EIRP_{90} adopted everywhere else here the same two become 3.2×10^5 and 1.3×10^4 . Nearly all of the difference is the product $N_{\star} \nu_{\text{rel}}$: this is a far smaller haystack than a wideband centimetre survey of thousands of stars, and the measure compares surveys rather than estimating occurrence rates.

6.3. What this experiment excludes, and what it does not

Excluded in a single epoch. Consider a continuously transmitting, unresolved carrier that lies outside the molecular-line mask, inside the searched frequency range and inside the linear-drift grid, and that is radiating while one of the archival blocks is on sky. Such a carrier would have produced a crossing nine times in ten above the EIRP_{90} of §5.1, a median of 1.6×10^{15} W per system on its best window, and no crossing of that kind survived toward any of the 60 systems the carrier search covers. **That is an exclusion of single-epoch detectability,**

and it holds for all of them. Four conditions bound it as sharply as power does: the coverage is sparse and inherited (Fig. 9), the median system covered over only 1.84 GHz; the drift grid reaches an acceleration that contains every known planet of the sample but one (Fig. 3); the statistic is built for emission confined to one 15.3–1953 kHz channel, anything broader being removed with the spectral baseline; and a carrier on for part of a block comes back at that fraction of its amplitude, at the cost §5.1 measures, while one absent throughout is not constrained at all.

Excluded through recurrence, over a far smaller domain. A crossing is not a detection (Table 2), and where the archive observed only once no power makes one. Recurrence could be tested at all for 39 of the 60 systems, and at a frequency the first epoch covered for fewer still (§5.6.1); **for the other 21 the confirmation completeness is zero.** Where it does apply it excludes a carrier holding its frequency inside the repeat window and radiating through both observations, at the amplitudes §5.6 quotes and no weaker ones.

Not excluded. Anything outside that domain; emission inside the $\pm 50 \text{ km s}^{-1}$ molecular-line mask at any strength; and any emission from the 17 566 stars within 40 pc that ALMA has not observed, which is almost all of them.

An illustrative engineering benchmark. A 12 m dish radiating 1 MW at 230 GHz with an aperture efficiency of 0.7 has a directive gain $G = \eta(\pi D \nu / c)^2 = 5.9 \times 10^8$ and so an EIRP of 5.9×10^{14} W, a depth this survey reaches for 16 of the 60 Class A systems. The same gain makes its beam $\Omega = 4\pi/G = 2.1 \times 10^{-8}$ sr, 34 arcsec across and illuminating 1.7×10^{-9} of the sky, so detection needs the Earth inside it. EIRP is not transmitter power — the same megawatt radiated isotropically would be 10^6 W — so these limits apply to beamed transmitters whose beam contains the Solar System and say almost nothing about omnidirectional beacons.

Two blind spots. The search is Stokes I only, so the discrimination polarisation would offer against unpolarised astrophysical emission is unavailable: ALMA’s feeds are linear and the circularity of a circularly polarised carrier lies in the cross-hands, which this analysis does not extract. And the frequency coverage is the archive’s and not a choice; none of the “magic frequency” arguments that have guided centimetre-band searches falls in an ALMA band.

6.4. What a dedicated ALMA technosignature experiment should do differently

A purpose-built experiment on the same telescope would be far stronger, and the changes it needs follow from the limitations above. *Channelise finely everywhere:* the 1249 coarse-channel windows cannot carry a carrier search at all. *Choose the targets and the frequencies:* an archive weighted toward dusty discs is the worst case for molecular-line confusion, the M dwarfs that dominate the nearby habitable-zone planet census are the least covered, and windows placed deliberately away from the bright transitions would recover most of the 3.05 GHz of the 47.7 GHz Class A union that the mask costs, 6.4 per cent of it — all three are decisions a proposal can simply reverse. *Repeat the epochs on pur-*

pose, at the same tuning: two or three visits separated by days and by years turn a crossing into a testable claim, and the separation is a scheduling constraint rather than extra sensitivity; Fig. 8 is what its absence costs. *Retain the visibilities:* rotating the phase centre to the star leaves a genuine point source entirely in the real part of the visibility. That asks whether the data contain a source *where the star is*, which is a stronger question than any rank against neighbouring positions in an image.

7. CONCLUSIONS

1. No published radio technosignature search above about 30 GHz has reached a star within 40 pc (§2), and an archive assembled for other science is at present the only way to reach one. The one ALMA band below those searched that holds any data on these stars is proprietary until 2027 and channelised far too coarsely to carry a carrier search.
2. The primary experiment is 402 fine-channel windows toward 60 stellar systems, covering 47.7 GHz of union bandwidth in disconnected intervals, searched for an unresolved carrier whose frequency drifts. A supplementary coarse-channel search for an unresolved spectral excess adds 1249 windows toward 81 systems and 90.5 GHz of union bandwidth, with no drift discrimination. Both come from one uniform reduction of 404 public execution blocks, 4.1 TB of raw data, and between them they span 90–873 GHz.
3. Sensitivity is a measured completeness of one epoch and not a threshold: EIRP_{90} is the power at which nine of ten carriers injected into the real calibrated visibilities come back through the unchanged trigger and localisation at the star, inside a single execution block. It is the power at which a carrier would have produced a crossing, not the power at which one would have been confirmed. The median system's best window reaches 1.6×10^{15} W, with a spread across systems of 6.8×10^{13} – 3.7×10^{16} W; carriers were injected into 49 of the 402 windows and the rest carry the class-average factor, whose measured dispersion moves that median by 2.5 per cent. One bias runs one way only and is outside that calibration: an injected carrier suffers no atmospheric decorrelation where a real one would, which leaves the median limit optimistic by 1 per cent and the windows above 400 GHz by 8 per cent.
4. All 56 threshold crossings are reported with their dispositions, four of them in the one execution block that fails the data-quality criterion of Appendix D.2 and reported apart from the 52 in quality-passing data. Of the 56, 21 fall within $\pm 50 \text{ km s}^{-1}$ of a catalogued transition at the star's own velocity, a coincidence that is reproducible and involves no judgement. Of those, three are with sulphur monoxide, reported and not acted on for a reason about the species; acting on them instead would make the split 21 to 35 rather than 18 to 38. The mask costs 6.4 per cent of the Class A union bandwidth, and re-searching the line-dominated windows with its channels removed adds no crossing. Chance, measured from each window's own control positions and channel grid, predicts 28.9 unattributed Class A crossings over the census against the 32 found there, or 30.8 with the

measured star-to-control transfer applied, and resampling the execution blocks allows 12–48. The count is consistent with chance, and the width of that range is the limit of the statement: an added population of as many as 19 crossings would sit inside it unremarked.

5. Every unattributed crossing was tested for recurrence in the star's own frame, over the whole of each further window covering the event frequency and at every drift searched, because an accelerating carrier need not return to one channel. None was found again: the largest statistic at any predicted cell is 4.71 against $T_{\text{trig}} = 5$. Of the 13 pairs of crossings lying close enough in frequency to be each other's repeat, the two fitted drifts and the elapsed time resolve the question for 7 of them, and a single carrier is excluded in every one; over the longer baselines they resolve none, and there the criterion tests the population rather than the pair. The same test accepts the one set of crossings that is present again, HD 285968's two carbon monoxide crossings. Of the 33 crossings that carry an exclusion, the repeat coverage rules out a carrier at a median 0.43 of the discovery amplitude with individual limits from 0.017 to 1.00, so the median is not a bound on the set; combining every covering block assumes a carrier that returns to one stellar-frame channel at the start of each, and where the repeat is shallow the criterion has little power and its silence is not an exclusion: 4 of these crossings are untested rather than excluded. What these tests exclude is a carrier that holds its frequency inside the window searched and radiates through both observations, and not a persistent transmitter of any kind.
6. What this survey could search and what it could confirm are different extents. Of the 60 systems the carrier search covered, 39 were observed in at least two independently separated epochs; for the other 21, observed once or only within a single day, confirmation was beyond reach at any sensitivity, and about those systems the null reported here says nothing at all.
7. Combining each surveyed star's epochs in its own rest frame deepens the flux limit by a median factor 1.52. For Proxima Centauri the stacked limit, measured by injection into the stack, reaches 2.2×10^{13} W over the primary census alone and 1.73×10^{13} W once the rest of the archive is added, in 15.625 MHz channels — a Class B limit on an unresolved excess, carrying no drift discrimination — for an emitter steady in the stellar frame to within one channel across 6.3 yr.
8. These results constrain unresolved carriers only within the domain actually searched (§6.3); what transfers beyond it is the obstacle, that channelisation chosen by other observers, and not collecting area, decides which kind of search is possible at all.

AUTHOR CONTRIBUTIONS

G.J.W. conceived the survey, defined the target sample and the selection criteria, and wrote the manuscript. R.D. built and operated the archival retrieval, calibration and search pipeline, implemented the spatial control statistic and the injection–recovery campaigns, and produced the released catalogue and its regeneration scripts. Both authors designed the vetting procedure, chose the

masked species, carried out the statistical validation reported in §4.1 and Appendix D, and agreed the dispositions of Table 9.

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COMPETING INTERESTS

The authors declare no competing interests. VBRL Holdings is a privately held company through which R.D. conducts independent research, named here solely as his affiliation; it had no role in the design of the survey, the selection of targets, the analysis, the decision to publish or the content of this paper, and provided no funding for the work.

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This paper uses ALMA data from the public ALMA Science Archive. The 404 primary-census execution blocks searched span 2013 October 6 to 2025 July 11 and belong to 65 ALMA proposals, whose codes, in the standard ADS/JAO.ALMA#*code* form, are recorded block by block in the data release together with each block’s epoch. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), NSTC and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia* (Gaia Collaboration et al. 2023), processed by the *Gaia* Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database (CDS, Strasbourg, France), of the NASA Exoplanet Archive (Akeson et al. 2013), and of the Splatalogue spectral-line database (Remijan et al. 2007).

DATA AVAILABILITY

The reference of record is a single permanent Zenodo deposit, cited by its DOI. It carries every number in this paper and the code that produced them, and contains: the 89-star target catalogue with its selection function; one row for each of the 1651 searched windows, carrying its execution-block identifier, frequency limits, native channel width, trial-drift grid, trigger statistic at the stellar position, control statistics of the 512-position ring and the window’s EIRP₉₀; the ALMA proposal code of every execution block, so that each of those rows resolves to the programme that took it, and each block’s epoch and barycentric velocity, measured from the delivered data wherever a measurement set survives; all 56 threshold crossings with their stellar-frame frequencies, line classifications and recurrence measurements, including for each the probability that a carrier at the discovery and at the trigger amplitude would have satisfied the confirmation criterion, and the catalogue’s own disposition column as computed at extraction, which this paper does not adopt — Table 9 is the authority for every disposition quoted here; the search statistic and the molecular-line list as files in their own right; the injection–recovery records; the block-by-block archival

verdicts and the ALMA delivery record for every block in scope that could not be searched; the diagnostic figures the printed paper refers to but does not reproduce; and the full pipeline with the commit that produced the deposited products. Two properties of the deposit are measured rather than claimed: 154 generated products regenerate byte-identically from the frozen inputs, and 41 headline quantities are re-derived from the released catalogue alone, without the pipeline.

Three choices that could create or remove an unattributed crossing carry an identifier checkable independently of the deposit. The calibration hold-out rule of Appendix B.1 and the block list its first guard depends on were committed to the project repository as `c75069040eab52ec1336376a4dae62114bb281a3` on 2026 September 14 at 07:10 UTC, 3 days before the first reserved block was searched. The molecular-line list is a single dated query of a public line catalogue (2026-10-06) deposited as a file, so it is reproducible rather than described, and it post-dates the fixing of the disposition criteria on 2026 September 11 (`f5f52e7804f1baad99515ea2d46ddd7e4e4cd546`), as does every species in it, so no species in the list was recognised late or early with respect to them. The repository at <https://github.com/Tilanthi/SETI> is a working mirror and may move on.

SOFTWARE

The pipeline depends on CASA 6.7.6 (casatools/casatasks 6.7.6-14; CASA Team et al. 2022), Python 3 with astropy (Astropy Collaboration et al. 2022), numpy (Harris et al. 2020) and scipy (Virtanen et al. 2020), and the ALMA Science Archive and SIMBAD TAP interfaces through astroquery (Ginsburg et al. 2019); all are free and open-source, with exact versions recorded in the frozen release.

APPENDIX

SPECTRAL RESPONSE, FREQUENCY CONVENTIONS AND THE PRIMARY-BEAM RESPONSE

The per-window catalogue, `catalogue/windows.csv` in the deposit, carries the columns every threshold quoted in this paper is checked from: the frame of each frequency, the instrumental response, and the primary-beam factor that turns a measured noise into a flux at the star.

Frequency frames, linewidth and smearing

Every frequency axis here — searched spectra, tabulated ranges and crossing frequencies alike — is *topocentric* as delivered by the ALMA correlator, and the search itself applies no velocity-frame shift. Velocities enter at the vetting stage, where the line mask is evaluated in each star’s own rest frame: it carries every crossing frequency to the barycentre at its own block’s epoch and then to the stellar frame before comparing it with a transition (Appendix C). The residual error is then the catalogue radial-velocity uncertainty alone, a few km s^{-1} within 40 pc; second-order relativistic terms at the 50 km s^{-1} scale are $\lesssim 5 \text{ kHz}$ at 350 GHz, below the finest native channel searched. The stacking is drift-aligned and inverse-variance weighted: coherent in frequency trajectory, not in electric-field phase.

The native channel width is also the linewidth convention for every EIRP threshold. The statistic is the single-channel flux density after drift correction, so $\text{EIRP}_{\text{trig}}$ is the minimum *total* power of a channel-confined carrier, flat in the assumed linewidth below the channel and never a spectral power density; flatness is exact only for a centred tone, and a boundary-straddling one divides its power between two channels, which is the loss the injections carry and one reason EIRP values are quoted to two significant figures. The Hz-resolution surveys’ limits are total powers in the same sense, each the best case of its own channelisation, which makes the axes of Fig. 1(a) commensurable. One number fixes the scale, illustratively and not achievably: rechannelising the median Class A window’s 488 kHz to the 2.79 Hz of a dedicated narrowband search would divide the threshold by 418, and no reprocessing recovers resolution the correlator never recorded.

Channel widths come from the measurement sets’ own `CHAN_WIDTH`; `MS v2` defines no “EFFECTIVE_BANDWIDTH” column, so the quoted channelisations are separations and are not claimed to be noise bandwidths.

ALMA applies online Hanning smoothing by default (Cortes et al. 2026), so the spectral response is ~ 2 channels wide and neighbouring channels are correlated. For the 0.25/0.5/0.25 kernel the coefficients are $\rho_1 = 2/3 = 0.667$ and $\rho_2 = 0.167$, with $\sigma_{\text{smoothed}}/\sigma_{\text{raw}} = 0.612$: the correlation is not confined to adjacent channels, the lag-two term being a sixth. The peak channel holds 0.50 of an unresolved carrier’s flux at a channel centre and 0.42 on a boundary, a median 0.48 over sub-channel phase, so a nominal threshold is optimistic by $\times 2.00$ – $\times 2.36$; the released catalogue carries the nominal threshold and both corrected values for every window.

The injected carriers are deposited with the correlator’s own channel response, at a sub-channel phase drawn afresh for each tone, so the smoothing loss sits inside the measured completeness rather than being applied to it. The deposition falls 8 per cent short of the ideal response averaged over phase, so the completeness is conservative by that much. Where in a channel a carrier was placed makes no difference to its recovered signal-to-noise, because a carrier at a typical searched drift rate crosses about 11 channels during one observation and so samples the whole response. Since the loss is carried end to end it must not be applied to the result as well: at the ninety per cent point of the recovery curve the peak channel holds $1.93\times$ the nominal trigger flux, which is that factor of about two, already spent.

Two measurements fix the smoothing state. The retained per-integration residuals of the second-stage local null have a lag-one spectral autocorrelation of 0.65 against the Hanning prediction 0.667, so the correlation is instrumental and those spectra are smoothed and un-averaged; and the archive’s reported effective spectral resolution is exactly twice the channel separation in 1505 of the 1651 windows, in both correlator modes; the other 146 report a smaller ratio, the signature of online channel averaging, where the peak-channel loss is smaller. Every released window has a reported resolution, so the blanket factor is right for most of both classes. *The correlation model is truncated at lag two:* the baseline step, a per-integration median over up to 65

channels, could imprint longer-range correlation that the released products retain no residuals to measure, which would make the independent-trials count of Appendix D an over-estimate.

Transmitters wider than one channel. The baseline filter (§4.1) is flat for features narrower than roughly half its effective width W and falls to zero above W , so the survey’s sensitivity is to channel-confined morphologies and W is the linewidth ceiling: 65–74 channels over the fine class and 30–59 over the coarse one, that is ≈ 32 MHz at 488 kHz and ≈ 0.9 GHz at 15.62 MHz. Within that ceiling a top-hat transmitter spanning N_{ch} native channels has per-channel flux the total divided by N_{ch} at unchanged noise, so its threshold is N_{ch} times the single-channel one.

Intra-integration smearing, which the injections do not model. A drift smeared by Δ channels within one integration keeps $\eta_{\text{smear}} = \text{sinc}(\pi\Delta/2)$ of the peak amplitude, so the effective threshold is $\text{EIRP}_{\text{trig}}/\eta_{\text{smear}}$. Each injected tone is deposited at one frequency per integration, so the campaign carries the channel response and the drift between integrations but not the sweep within one, and this term is evaluated instead of measured. The integration length survives for 1046 of the 1651 windows, 451 of them in the released catalogue and 595 more in the exposure records the search kept; over those the excursion reaches 1.10 channels at the top of a window’s drift range, and η_{smear} falls below 0.99 in 13 of them, on 6 stars: HIP 10679, HD 172555, β Pic, η Crv, CP–72 2713 and HD 48370. Their thresholds rise by $\times 1.02$ to $\times 1.74$, and the worst compounded with the worst Hanning factor, $\times 2.36$, gives $\times 4.1$. For the other 605 windows the length is not recorded and the term is bounded instead: at 11.6 s, the longest integration delivered anywhere in the campaign, none of them falls below 0.996, so no window is left unaccounted for. The factors above are ceilings, taken at the fastest drift each window searched; averaged over the drift range the largest is $\times 1.18$. The limits quoted in this paper do not carry the factor, because the released catalogue does not: it deposits each window’s own threshold, and this term is not one of its columns, so applying it here would put 13 limits one step ahead of the data behind them. Applied, it would leave the per-system median of Fig. 1(a) where it is and take one system out of the 25 that reach 10^{15} W, leaving 24.

The primary-beam response, and the two columns that let a reader recompute it

The sensitivity quoted for a window is $S_{\text{min}} = 5\sigma/A$, where A is the primary-beam response at the star’s offset from the phase centre: the noise σ is measured in apparent flux, the limit is quoted in true flux, and the two differ by A . That offset is not a nuisance. Archival fields are pointed at a catalogue position, often decades old, while these are high-proper-motion stars, so the star can sit well off axis: over the 1651 released windows the offset has a median of $0.089''$, a 99th percentile of $7.43''$ and a maximum of $8.13''$, and A runs from 0.999984 in the median down to 0.5522, below 0.99 on 91 windows and below 0.9 on 34. *None* falls below one half, and every one of the low-response windows is a physically explicable J2000 pointing on a fast-moving star rather than an error. A window is kept only where the response is at least 0.5, a correction of at most $\times 2$, since at low response

a small error in the beam model becomes a large error in inferred flux. The offsets are bimodal and the floor sits in the gap: the last retained window has response 0.55 at $0.46\theta_{\text{PB}}$, the first window beyond it 0.35 at $0.62\theta_{\text{PB}}$, and nothing lies between, so any floor between those two responses retains the same 1651 windows and the value chosen does not matter. Beyond it lie the 12 ϵ Eridani Band 6 windows, which were withheld as a band and not window by window. 8 of them, in 2 blocks, need a correction of up to $\times 3.5$ — past the first sidelobe transition, where the Gaussian form fails — and so fail the floor outright. The other 4, in block Xb9c01c_X2004, place the star at the pointing centre and need no correction at all, so the floor is not what removed them; and at 3.22 pc the star is too near to be left without a limit, so theirs are given here. They carry 0.10 window-hours and no threshold crossing, the largest statistic being 3.8 where the trigger is five. The one fine-channel window, at 488 kHz, reaches 1.8×10^{14} W and the 3 coarse ones at 15.6 MHz 9.3×10^{14} – 1.1×10^{15} W; none enters a census statistic. The flux-scale comparison of §4 is a separate continuum measurement on the same star: a block mean over the 4 compact-array executions that place it at the pointing centre, 0.22 h in Band 6, where no beam correction enters either.

The correction is applied exactly once. The offset is computed for every measurement set from its own FIELD:PHASE_DIR and the propagated stellar position, so no window carries a zero-by-assumption offset, and it enters in one place only: $S_{\text{min}} = 5\sigma/A$ closes on all 1651 released rows to $8.7\text{e-}05$ in relative terms. The count is in two parts, because on 10 rows whose products are not retained A is back-solved from $5\sigma/S_{\text{min}}$ and flagged as such, so there the closure is arithmetic; the audit is a test on the other 1641 rows, where A comes from the search's own stored geometry and closes to $8.7\text{e-}05$. Nothing else needs correcting: the statistic of Eq. 1, the control ranks and the attribution are *ratios* formed inside one window, in which A divides out identically.

The response applied is pb_atten: a Gaussian of full width $1.22\lambda/D$ — an Airy first-null coefficient — at each window's *own* dish diameter, reproduced over all 1651 released rows to 2.6×10^{-5} against 7.7×10^{-2} for the same Gaussian forced to $D = 12$ m. The deposited `theta_pb_arcsec` is that width at $D = 12$ m for every row, including the 1054 taken on the $1.7\times$ smaller compact-array dishes, so A must be read from `pb_atten` and cannot be recomputed from it.

The beamwidth convention is a bounded systematic, and the release carries both conventions so that a reader need not take ours. ALMA's primary-beam full width is $1.13\lambda/D$, narrower than the first-null coefficient, so it attenuates more and makes every limit *shallower*: by nothing at all in the median window, by more than one per cent in 40 of 1651 and by more than ten in 4, the largest being $+10.3$ per cent — those few all LP 736-15 at $8.1''$ off axis. The lowest response in the release moves 0.5522 to 0.5005, still clear of the 0.5 retention floor but by a margin worth saying, and `pb_atten_fwhm` in the deposit carries that column window by window. No disposition changes with the convention. The remaining model uncertainty is also bounded by measurement: CASA's blocked-Airy response is better physics than either Gaussian, the largest ratio to it over the whole re-

lease is 1.0745, and the counts of systems reaching 10^{15} W are identical under any of the three.

The star's apparent direction includes the annual parallax, and the amplitude a compact source retains when that is left out of the assumed phase centre is measured rather than modelled. The displacement decides whether the loss can be treated as a scalar at all: a small fraction of a synthesised beam attenuates a compact source and does nothing else, while a displacement approaching a beam puts the phase wrong on the longest baselines. Re-extracting the 8 windows of 2 blocks furthest out at the full apparent position — proper motion, parallax and radial velocity where catalogued — leaves each window's noise unchanged to 0.20 per cent and alters no disposition. It cannot test the amplitude factor, because that factor describes a source and those windows hold none: the statistic is a maximum over a plane of drift trajectories, so moving the extraction point by two thirds of a beam draws a fresh noise realisation, and the ratio scatters from 0.78 to 1.15 about a median of 0.98 instead of approaching the 0.49 a source would give. A carrier injected at the apparent position and recovered from the *same* visibilities at both positions does return the normalised dirty beam at the omitted displacement, measured on that block's own baselines with no model in it: 0.92 against a predicted 0.89 at 0.25 of a beam, and 0.58 against 0.49 at 0.67. That it exceeds the model at the larger displacement is the search's doing, since maximising over a plane of trajectories lets a partly decohered carrier still find a cell, so a limit corrected by the model is shallower than it needs to be.

The injection campaign is blind to A by construction, and the bias that follows is exactly zero. The injector deposits an unattenuated amplitude at the star's own direction cosines and refers the ladder to the *apparent* 5σ on all 40 scored units, so an injected apparent amplitude recovered nine times in ten implies a true flux larger by $1/A$ while P_{trig} is the power of a true $5\sigma/A$ and A cancels identically; 4 of the units have apparent and corrected S_{min} differing by more than a part in a thousand, so the cancellation is tested rather than vacuous. Being self-consistent in apparent flux, the campaign cannot falsify the beam model; and it reaches only $A \geq 0.9102$, whereas 75 windows on 5 stars lie further off axis, down to $A = 0.552$, so a limit quoted for one of those rests on an extrapolated curve; their 6 crossings are ratios and unaffected.

A few rows carry a catalogued noise that no retained product reproduces, and they are declared rather than reconciled: on 4 windows of HD 285968 in block A002_Xd98580_X3bf6 the catalogued σ is 0.88–1.32 per cent *higher* than the surviving product's, so the limits there are weaker and not stronger. With the 10 windows whose products are not retained the declared provenance gap is 14 windows; the attenuation columns are unaffected.

THE SPATIAL RANK, AND WHAT IT CAN SUPPORT

The spatial rank asks whether an excess sits at the star or anywhere else in the field. It is a diagnostic, no disposition rests on it, and this appendix is the measurement that settles why: The star-versus-control rank is an empirically useful way of deciding which crossing to examine

TABLE 7

THE BLOCK AND WINDOW LEDGER. EVERY SET OF EXECUTION BLOCKS USED ANYWHERE IN THIS PAPER, WITH THE RESULTS IT ENTERS; THE ROWS SUM, AND EACH INDENTED GROUP SUMS TO THE ROW ABOVE IT. THE PRIMARY CENSUS IS THE SURVEY: UNLESS A STATEMENT NAMES ONE OF THE OTHER SETS, IT IS COMPUTED OVER THAT ROW. HOURS ARE WINDOW-HOURS: ON-SOURCE TIME SUMMED OVER SPECTRAL WINDOWS, SO A BLOCK DELIVERING SEVERAL WINDOWS AT ONCE CONTRIBUTES ITS ON-SOURCE TIME ONCE PER WINDOW. THE CLASSES THEREFORE PARTITION THE HOURS AS THEY PARTITION THE WINDOWS. THE TELESCOPE ON-SOURCE TIME BEHIND THE PRIMARY CENSUS IS 280 h, OF WHICH 152 h CARRIES AT LEAST ONE CLASS A WINDOW AND 276 h AT LEAST ONE CLASS B WINDOW — FIGURES THAT OVERLAP AND MUST NOT BE ADDED. CONFIRMED PLANET COUNTS, HERE AND IN THE DEPOSITED TARGET LIST, ARE TAKEN FROM THE NASA EXOPLANET ARCHIVE **PSCOMP** TABLE (AKESON ET AL. 2013), ACCESSED 2026 OCTOBER 4.

Set of execution blocks	Blocks	Windows	Stars	Systems	Hours	Results it enters
Archival scope: blocks the query exposes	656	–	–	–	–	defines the scope; the sum of the next two rows
... processed with the frozen pipeline	479	–	–	–	–	the second panel below
... not processed at the freeze	177	–	–	–	–	the third panel below
Processed with the frozen pipeline ^c	484	–	–	–	–	the sum of the three rows below
Primary census	404	1651	89	82	1094	every limit, the crossing list, every chance expectation
Class A, fine channels ^a	–	402	65	60	227	the drifting-carrier experiment
Class B, coarse channels ^a	–	1249	88	81	867	the spectral-excess experiment
Reserved hold-out	77	315	36	–	214	the rank displacement, the false-alarm tail factor, the radial control profile
Withheld: primary-beam form invalid ^d	2	8	–	–	–	nothing; withheld
ε Eri at the pointing centre ^c	1	4	1	1	0.1	reported limits only; no census statistic
Not processed at the freeze: what became of each	177	–	–	–	–	the sum of the five fates below
Calibrated and searched	144	–	–	–	–	the sum of the four rows below
Archival extension , reported	115	463	16	16	269	recurrence, and the null on data the frozen pipeline had not seen
Searched after the extension was frozen	8	32	–	–	–	nothing; later than the reported extension
Withheld: primary-beam form invalid ^d	8	32	–	–	–	nothing; withheld
No window reached the export	13	0	–	–	–	nothing
No calibration in the ALMA delivery	18	–	–	–	–	nothing; the archival gap
Calibration failed	11	–	–	–	–	nothing; the archival gap
Not attempted	3	–	–	–	–	nothing; the archival gap
Working disk insufficient	1	–	–	–	–	nothing; the archival gap
Beyond 40 pc: out-of-sample control	47	198	22	–	88	the external null; no science count
External calibration sample ^b	326	1322	56	–	–	the rank calibration and the positive control; no science count

^aThe two classes partition the census windows, not its blocks: 252 blocks carry a Class A window and 389 a Class B one, and most carry both, so the hours of the two rows partition the windows and not the observing time.

^bNot an out-of-sample set, and not an addition to this ledger: every one of its 326 blocks is counted in a row above, 263 in the primary census and 63 in the hold-out, so it is listed apart only to keep it from being counted twice. The harvest it was drawn from holds one further block, toward a star outside the 40 pc work list, and that block is excluded here.

^cThis panel and the scope panel above overlap rather than add: 479 of these 484 blocks are the in-scope row above, and the other 5 carry no member observing unit in the archive metadata, so the scope query does not expose them. 3 of the 5 are in the primary census.

^dEvery one is an ε Eri Band 6 block, and the band was withheld as a whole: in most of these blocks the star sits far enough off axis (response 0.28–0.35, below the 0.5 floor) that the Gaussian primary-beam form is not valid.

^eThe windows of the same band that place the star at the pointing centre, at a response of 1.00, and so pass the floor. They were set aside with the rest of the band; their limits are in Appendix A and they enter no census statistic. The archival-extension row above holds a further 16 such windows, which enter nothing either.

first, but the stellar and control positions are not demonstrably exchangeable under the null. The rank therefore has no calibrated per-window false-alarm probability and is not used as evidence for detection.

The star lies at the phase centre, inside the control annulus, and the asymmetry is real. *It is not a primary-beam effect.* Spectra are extracted from phase-rotated visibilities in apparent flux and are never corrected for the primary-beam response (Appendix A), so the familiar rise of image noise with distance from the pointing centre cannot operate here; and were it operating, it would make the control probes noisier than the star and place the stellar rank *above* one half, the opposite of what is measured. No mechanism for the displacement is established. What the control ensemble does show is a weak rise of its own level with radius, and the obvious repair is to remove it: each probe is standardised within its own radius bin, the radii being known from the fixed draw, with no free parameter.

Applied out of sample, the correction does not repair the statistic. Frozen — probe radii, bin edges and stan-

dardisation alike — and applied unmodified to 778 windows that played no part in its construction, it moves the median stellar rank from 0.409 to 0.407 against the exchangeable one half, further from it, in both out-of-sample partitions separately. Rescoring the whole survey with the gradient removed moves the list of crossings out-ranking all their controls from 12 windows to 11, losing 2 and gaining 1, and leaves the survey median rank slightly further from one half than before.

The radial gradient is real, but removing it leaves the median no closer to one half, so the residual is not radial and neither normalisation yields a calibrated tail probability. We therefore keep the frozen rank, read it only as an indication of which crossing to examine first, and rest every disposition on physical evidence or on recurrence. The radial correction, the region-maximum audit and the pseudo-star tests each leave every disposition unchanged, and because the rank gates nothing, none of this enters a sensitivity limit.

One mechanism can be excluded by measurement, and it is the one that would matter most. Equation 1 divides

every position by a single noise scale shared by the star and all its probes. A star bright enough to carry multiplicative calibration residuals of its own would then have a noise the shared scale does not describe, and its statistic would be inflated with nothing in the search to reveal it. The ratio of the two noises is measurable from the released control vectors alone: because the star and each probe are divided by the same scale and maximised over the same channel $\times$ drift grid, the stellar statistic divided by the median of the 512 control statistics *is* that ratio, and it is one when the two noises agree. Over the 1595 windows carrying no threshold crossing its median is 1.001 (star-clustered 95 per cent 0.994–1.004), and it does not climb with the brightness of the field, which is what a multiplicative error would scale with: 1.002 in the faintest fifth of control rings against 0.999 in the brightest, and 1.09 over the 8 windows whose ring exceeds 14σ and is therefore set by resolved emission rather than noise. That the ratio *can* rise is shown by the 56 windows that do carry a crossing, where it is 1.18. Part of the reason it does not rise otherwise is structural: a gain error multiplying a continuum raises every channel of an integration by the same factor, and the per-integration median along frequency removes exactly that. **A stellar-position noise denominator would therefore move no statistic in this survey.**

The reserved hold-out, and what it calibrates

The statistic and the mask were settled on the searched sample and cannot also calibrate themselves, so a hold-out was reserved by a rule fixed and recorded before any reserved block was searched: a block is held out if and only if $\text{sha256}(\text{UID})_{[:8]} \bmod 5 = 0$, subject to two guards that use identifiers alone — a block on the pre-registered list deposited with this paper stays in the survey, and no system loses every block — so that the assignment can be recomputed from the UIDs and that list alone. The statistic, the trigger and the recurrence criterion were all fixed before the rule itself, and the first reserved block was searched 3 days later, so every diagnostic built afterwards is incapable of promoting a window. Of 484 blocks processed, 77 are reserved (15.9 per cent), giving 315 windows toward 36 stars with no star and no system lost from the headline sample. Three quantities are calibrated on those blocks and on nothing else.

First, the rank displacement. Out of sample the stellar rank has median 0.405 where one half is expected, with a block-clustered 95 per cent interval of 0.341–0.458 over 315 windows, and 14 per cent of ranks below 0.1 against 6 per cent above 0.9, so the distribution is displaced rather than growing a tail. The displacement runs the way that matters: a rank below one half places the star high, and the star in fact outranks all 512 probes in 14 of the survey’s windows against 3.2 if the positions were interchangeable. The direction therefore inflates star-first outcomes rather than suppressing them, which is why the chance expectations of Appendix D are calibrated on the measured rate. Clustering removes neither the shift nor its size: 99 of 149 blocks of the epoch-extension campaign have a mean rank below 0.5, and a star-clustered bootstrap places the median at 0.440 (95 per cent 0.42–0.46). The searched sample leans the same way but by less (median 0.49, star-clustered 95 per cent 0.47–0.51), and the two intervals do not overlap, so the displacement

is a property of these reserved blocks and is not reproduced in sample. Either way it runs in the direction that inflates star-first outcomes, which is the quantitative reason the rank is read as a diagnostic and never as a significance.

Second, the false-alarm tail factor — the measured rate at which a no-signal position ranks first, divided by the rate exchangeability would give. It is 1.2 on 5 040 trials, with a bootstrap clustered over the 77 reserved blocks giving 0.6–1.9. It is a statement about the rank and not about the trigger, so it is reported as a diagnostic; the quantity the chance expectations need is the exceedance rate at the stellar position, measured on the same blocks in Appendix D.

Third, the radial behaviour of the control ensemble: the standardised level runs +0.12, +0.10, +0.12, +0.16 across the annulus, a mean per-window slope of $+0.067 \pm 0.033$, small and of the wrong sign to account for the displacement.

The *external calibration sample* must not be confused with the hold-out, and it is not an out-of-sample set. Its 326 execution blocks were reserved to calibrate the rank and to supply repeats the hold-out does not have, and 263 of them are primary-census blocks and 63 hold-out blocks, with 0 outside both: it enters no count of the science sample because its blocks are already counted there, not because it is external. Over its 1322 windows the frozen pipeline returns 7 crossings outranking all their controls — 4 of them β Pictoris CO — and 3 unattributed against 3.7 expected at the measured tail rate.

One crossing through the whole chain

CP–72 2713 is the clearest worked example, because every step returns a verdict rather than a shrug. In Band 7 the crossing reaches $T_\star = 5.809$ in channel 35 of its window; it is unattributed, with the one nearby transition and its provenance discussed in Appendix C; both diagnostics favour it, in that it outranks every one of the 512 control positions and the drift-following visibility fit puts the emission at the stellar position, $\text{Re}/\sigma = 5.1027$ against a control-position maximum of 1.8222; and neither of those is a promotion (§5.5).

What disposes of it is the repeat coverage. One of its 5 covering blocks matches the discovery in tuning, channel width, channel count, integration count and on-source time, registers to 0.18 of a channel, and is the deeper of the two: a carrier radiating at the discovery amplitude through both observations would have returned $T_{\text{pers}} = 6.21$ there, and that block alone puts the pair 4.6σ apart. Combining all 5 gives $T_{\text{pers}} = 6.30$ and 4.8σ , and rules out a carrier above 0.66 of the discovery amplitude. The nearest covering block is 2.05 h away, but 4 of its 5 covering blocks are more than a day away, the nearest 2.05 h: it excludes a carrier of that strength radiating through both epochs and says nothing about an intermittent one. **The disposition rests on the recurrence test and on nothing else**, which is the chain applied to every crossing in the survey.

What the recurrence statistic reads when nothing is there

The exclusion in Table 9 is one formula. Writing S and σ_d for the discovery amplitude and its error, and $\bar{a}$ and

$\bar{\sigma} = (\sum_i \sigma_i^{-2})^{-1/2}$ for the inverse-variance combination of the repeat readings at the predicted channel,

$$\sigma_{\text{excl}} = \frac{S - \bar{a}}{\sqrt{\sigma_d^2 + \bar{\sigma}^2}} = \frac{T_{\text{pers}} - T_{\text{rep}}}{\sqrt{1 + (T_{\text{pers}}/T_{\star})^2}}, \quad (\text{B1})$$

with $T_{\text{pers}} = S/\bar{\sigma}$ and $T_{\text{rep}} = \bar{a}/\bar{\sigma}$, so every exclusion follows from three columns of that table and nothing else. T_{rep} is read at the carried drift and at one cell; the largest value over every drift trial at the same cell is tabulated beside it, and because it is a maximum over of order 10^2 trials it is positive on average whatever is there and is never the quantity T_{pers} is compared with. Because the discovery amplitude carries its own error, no exclusion can exceed the strength of the discovery it is built from — and that is also why it is not the strongest statement available. An amplitude selected as the largest of 10^2 to 10^6 cells at $T_{\star} \simeq 5$ is biased high, so a real carrier would be fainter than measured, and the quantity that survives the bias is the limit $(T_{\text{rep}} + 5)/T_{\text{pers}}$: the fraction of the discovery amplitude still ruled out at the trigger.

The same three numbers say where the test has power and where it has none. The criterion asks the repeat to reach the trigger, and the reading there is centred on zero with unit scatter, so a carrier radiating through both observations at the discovery amplitude satisfies it with probability $\Phi(T_{\text{pers}} - 5)$, and at the trigger amplitude $5S/T_{\star}$ — the faintest carrier that could have produced the crossing at all — with $\Phi(5T_{\text{pers}}/T_{\star} - 5)$. Both follow from columns Table 9 already prints, both are tabulated for the two events of Table 5, and both are deposited for every crossing. Where they are small the crossing is untested rather than excluded, and it is reported that way.

The statistic read at the predicted channel is centred on zero and scatters by one, measured and not merely designed. Over 511 single-cell readings of every repeat block in the census and the hold-out its mean is $+0.014$ and its scatter 1.046 ; at the neighbouring channels of the 2 crossings whose repeat spectra survive — random cells, same registration, same drift — $+0.071$ and 0.966 over 638 cells, with 3.19 the largest value anywhere within $\pm 15 \text{ km s}^{-1}$ of their predicted channels; and at the control positions of the discovery windows -0.036 and 1.034 . The largest of the 8 retained control probes of each repeat block averages 1.449 against the 1.424 expected for a maximum of 8, which is the point: **a maximum is positive on average whatever is there.** The largest of the n_{rep} per-block readings averages $+1.102$ over the crossings carrying exclusions, where the null above predicts $+1.130$ for the same n_{rep} , while the combined T_{rep} of Eq. B1 averages -0.119 . Widening the window costs trials and not calibration: the exceedances of §5.6 match what the same accounting predicts, and HD 69830's windows are the only ones too narrow for the excursion their crossing's drift implies.

THE LINE MASK

A crossing that falls on a known molecular transition has an astrophysical explanation to hand, and the mask exists to say which crossings those are. It reports a frequency coincidence first and an astrophysical judgement about the coinciding species separately, so that the reproducible part can be recomputed and the arguable part

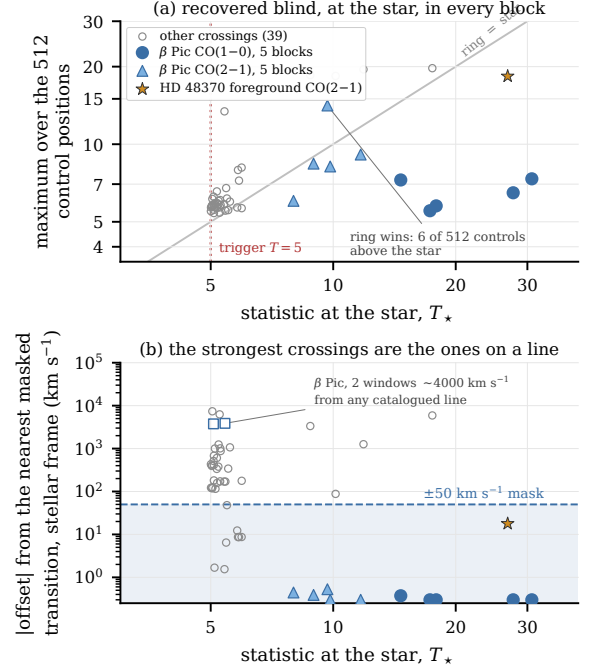


FIG. 10.— The β Pictoris positive control, in both directions. (a) The spatial screen. One point per execution block in which the blind search recovers circumstellar carbon monoxide, plotted as the stellar statistic against the maximum over that window's 512 control positions: the 9 blocks in which the star outranks every control, and one further block in which the ring wins. *The control is that the search recovers a real narrow line at a stellar position, in two transitions and in every one of these blocks.* It is not that the star outranks the ring everywhere. In block X7116f1_X1785, Band 6, the control ring reaches 14.13 against 9.68 at the star, with 6 of the 512 control positions above it, so the screen rejects a window holding carbon monoxide that is real, centred on the star and independently mapped: **the screen measures compactness and not authenticity** (§5.4). (b) Line confusion, which is what makes the screen's job hard. Each crossing is plotted by its velocity offset from the nearest masked transition *in its own star's rest frame*, with the adopted $\pm 50 \text{ km s}^{-1}$ half-width marked. The 10 β Pictoris carbon-monoxide crossings land $0.1\text{--}0.5 \text{ km s}^{-1}$ from CO, at the resolution of the systemic velocity itself, while the two CS(5 $\rightarrow$ 4) crossings of the same star sit $3765\text{--}3870 \text{ km s}^{-1}$ from anything catalogued and are not attributed at any width. 39 further crossings are drawn in grey, with HD 48370 marked; 52 of the 56 are plotted, the remainder being crossings for which the release stores no control ring. *A crossing is identified by where it falls in this plane and not by how strong it is*, which is why the line mask and not the trigger is the productive test.

argued with.

The list, and when it was made. The mask holds every transition of the 12 species these spectral setups cover — the three CO isotopologues, HCN, HCO⁺, CS, CN, SiO, H₂CO, SO, [C I] and hydrogen recombination — that falls inside one of the 46 searched frequency intervals with an upper-state energy below 150 K. That rule is the whole of it, and it admits 111 transitions (Table 8). No strength cut is applied across species, since any threshold admitting the strong lines of HCN would discard the brightest lines in these fields; within a species the main hyperfine components are kept and the weak satellites dropped. The list is a single Splatalogue query of CDMS and JPL (Pickett et al. 1998; Müller et al. 2005) made on 2026 October 6, after the disposition criteria were fixed on 2026 September 11, and frozen with the data release, so it is reproducible as a file rather than as a description. It is built from what the setups cover and not from what each

TABLE 8

THE LINE MASK, BY SPECIES: THE NUMBER OF TRANSITIONS N IT HOLDS, THE FREQUENCY RANGE THEY SPAN INSIDE THE SEARCHED INTERVALS AND THEIR UPPER-STATE ENERGIES. HYDROGEN RECOMBINATION LINES CARRY NO TABULATED E_u . THE TRANSITION-BY-TRANSITION LIST IS IN THE DATA RELEASE.

Species	N	ν_0 (GHz)	E_u (K)
H	5	92.03–231.90	
SO	15	100.03–682.68	16–143
H ₂ CO	13	101.33–491.97	10–106
C ¹⁸ O	3	109.78–329.33	5–32
CN	50	113.12–680.30	5–114
CO	3	115.27–345.80	6–33
CS	3	195.95–342.88	24–66
SiO	2	217.10–347.33	31–75
¹³ CO	8	220.40–330.59	16–32
HCN	7	354.50–354.51	43–43
HCO ⁺	1	356.73–356.73	43
[C I]	1	492.16–492.16	24

system is thought able to carry.

The frame is the star’s own, and the sign is not the usual one. A crossing’s frequency is carried from the topocentric value to the barycentre using its own block’s pointing and mid-time, and then to the stellar rest frame using the star’s systemic velocity. The barycentric term alone spans 58 km s^{-1} across the survey, so a tolerance applied in the observed frame would spend most of its budget on the Earth’s motion. The 3 HD 285968 epochs show the transform working in the only way that can be checked: 12 km s^{-1} apart as observed, they agree to 0.03 km s^{-1} in the star’s frame. One row looks as though no transform had been applied, and the reason is arithmetic: toward HD 139084 B the barycentric term is $+4.5161 \text{ km s}^{-1}$ and the systemic velocity $+4.5500 \text{ km s}^{-1}$, so each moves the frequency by 5.2 MHz and the two nearly cancel, leaving 39 kHz ; its coincidence, $+24.4 \text{ km s}^{-1}$ from CO(3–2), is unaffected. Throughout, $\Delta\nu_\star = c(\nu_\star - \nu_0)/\nu_0$, so a positive offset means the crossing lies *above* the transition in frequency — the opposite sign to the velocity convention of a spectral line.

The coincidences, and the one species we do not read as emission. Of the 56 crossings, 21 fall within $\pm 50 \text{ km s}^{-1}$ of a transition of the list in their own star’s frame; that count is the first layer and all the velocity criterion decides. Coincidences with sulphur monoxide account for 3 — CD–57 1054, CP–72 2713 and HD 161868, at -1.7 , -12.3 and -48.0 km s^{-1} from transitions of upper-state energy 21–87 K, each inside a Band 7 window that also covers CO(3 $\rightarrow$ 2) — and two things argue against reading those as emission. Sulphur monoxide has not been reported in a debris disc, where the second-generation gas is CO, C and O; and toward CP–72 2713 the CO(3 $\rightarrow$ 2) line falls in the same window and is absent to 10.5 mJy in a 0.42 km s^{-1} channel while the crossing itself is 12.2 mJy , so an SO line there would have to outshine an undetected CO(3 $\rightarrow$ 2) line in the same data. Both arguments are about the species and neither about any one crossing, so all 3 are treated alike: the offset is printed beside each in Table 9, where a dagger marks the rows this applies to, and disposes of none of them, giving 18 attributed against 38 unattributed, of which two outrank every control position in their own windows. Reading them as attributions instead gives 21 and 35, and 1 such crossing; the recurrence test excludes a persisting carrier for all 3 either way, at 4.1 – 5.5σ , so the choice moves

counts and no conclusion. The reserved hold-out holds 2 coincidences, 1 of them the same species and treated the same way, leaving 1 attributed and 7 unattributed there (§5.6.2).

The half-width, and what it costs. A one-channel test is too tight for circumstellar gas, whose emission is shifted by many channels: the width follows from Keplerian motion where these molecules survive, $\lesssim 13 \text{ km s}^{-1}$ beyond 10 au, and is $\pm 50 \text{ km s}^{-1}$. At that width the mask removes 3.05 GHz of the 47.7 GHz Class A union, 6.4 per cent of it, against 5.04 GHz of the 118.1 GHz both classes cover, 4.3 per cent: the cost falls on the primary experiment, because its windows are the line-tuned ones. Between ± 20 and $\pm 100 \text{ km s}^{-1}$ the mask labels 17 to 19 of the crossings, and the 2 that outrank all 512 control positions of their own windows are the same two at every width.

A bright line costs its whole window, and the windows were therefore searched again. Because the search records one statistic per window, a weaker carrier in a window whose largest value is a molecular line could not have been recorded at all (§5.3), so the price of line confusion is the window and not the masked channels in it: 21 Class A windows spanning 6.00 GHz of unique frequency, 12.6 per cent of the Class A union. Most of that is recoverable, because the per-channel statistic those maxima were taken from is retained: removing the masked channels and taking the largest value of what is left researches 18 of them, leaving only the 3 fine windows that are narrower than the mask itself and whose 0.108 GHz, 0.23 per cent of the Class A union, is lost outright. The surviving maxima run 4.13 to 5.04, so one of them reaches the trigger: β Pic at 115.6200 GHz, $+927 \text{ km s}^{-1}$ from the nearest transition. It is what noise gives at this threshold, since 12 of the 144 control positions of the same windows reach it outside the mask as well, in 8 of them; and it does not recur, the 4 other blocks covering that cell reading $T_\star = 1.7$ to 3.4.

The labelled crossings are tested for recurrence too. Of the 18 attributed crossings, 15 have a later block covering the predicted cell and 13 are present again, toward β Pic and HD 285968 at $T_\star = 5.9$ to 30.8 — the test working in the positive direction on emission that is certainly real, and no exclusion is formed for those rows, because what it would exclude has just been observed. For the 2 not recovered a repeat window’s own largest statistic bounds the predicted cell from above, and Table 9 prints the row as a bound; 3 have no covering block at all (HD 139084 B, HD 48370 and HIP 10679).

Image-sideband leakage accounts for no crossing. The mask holds signal-sideband frequencies, and the 2 unattributed crossings that share a block with a CO(2 $\rightarrow$ 1) crossing in the opposite sideband would need a first local oscillator outside the range their own block’s spectral windows permit, 235.02–235.68 GHz; their image frequencies fall at least 1353 km s^{-1} from every masked transition.

A longer list is not a better one. Every Splatalogue transition over these intervals with an Einstein coefficient above 10^{-5} s^{-1} — 5,114 transitions of 235 species — would mask 75.1 per cent of the union and move 27 crossings across the line, one of them rank-leading. At these frequencies almost anything lies within $\pm 50 \text{ km s}^{-1}$ of something in such a list, which is why the species are

restricted to what these setups cover and why the outcome is a label rather than a verdict.

THE FALSE-ALARM EXPECTATION AND THE CONTROL ENSEMBLE

This appendix gives the chance expectation for the unattributed crossings, what it is conditioned on, and what the control ensemble calibrating it can support.

What the expectation is conditioned on

The expectation is built from the control positions, the only empirical null the data provide, and each Class A window contributes two quantities measured on itself. The fraction of its own 512 control positions that reaches the trigger is the probability that a position in that field does so — no cell count enters it and no form is assumed for the tail — and summed over the windows 36.1 are expected to trigger by chance. The share of its own channel grid lying inside an attribution window, in that star’s rest frame, is the probability that such a position would be attributed, a control maximum being free to fall on any channel. Weighted by the first that share is 20 per cent, against a median of 6.0 per cent per window, because 15 narrow line-tuned windows lie wholly inside the mask and their fields, which hold real emission, are where a control position most often triggers. What is left is 28.9 expected unattributed Class A crossings against 32 observed.

Both sides run over the 402 Class A windows of the released catalogue, every one of which it tabulates window by window, so the expectation is a sum over rows a reader holds and nothing is carried at an assumed rate. A further 15 Class A windows arrived with the archival tail and lie outside it, together with the two unattributed crossings toward HD 14055 they hold, which are reported beside the comparison and enter neither side of it. A systemic velocity is measured for the stars of 277 of the census windows and the others are taken at rest, which moves the masked share by 3.0 points. No window holds more than one crossing, so both sides count windows.

The one assumption the searched data cannot test is that a star behaves like a control position in its own field, and the reserved blocks measure it in the quantity used here. Their stars reach the trigger in 8 of 62 fine-channel windows where the control positions of those same windows predict 7.5: a ratio of 1.07, with 0.48–1.79 allowed by resampling the reserved blocks, and no two of the events in one block. In the 253 coarse-channel reserved windows no star reaches the trigger, against 0.04 predicted. The same blocks place the stellar rank below one half (Appendix B), which is the same departure read in the body of the distribution rather than in its tail and runs the same way. Carrying the measured ratio through to the expectation gives 30.8 against the 28.9 above, and the observed 32 sits inside the range either way; the ratio is itself consistent with unity, so the uncorrected sum is the one quoted and this is the size of the correction it leaves out.

The 12–48 quoted beside the expectation is what 20 000 resamplings of the execution blocks allow, each window’s rate redrawn from its own 512 probes: windows share blocks and are not independent trials, and the ensemble is finite. Those two terms alone give 26–32, so what widens the range is the calibration above. Read as the

interval it is, it leaves the expectation anywhere between 12 and 48: an added population of as many as 19 events would be absorbed by that width, and that, rather than any smaller figure, is the limit on what the comparison can exclude.

Every Class B crossing in the survey comes from the one execution block of §D.2, and setting it aside the coarse class adds almost nothing to either side: over both classes together 29.0 are expected against the same 32 observed. Only 455 of the 1651 windows hold any control position above $T_{\text{trig}} = 5$ at all, a coarse single pointing having too few cells for its own field to reach the trigger by chance, so the Class A comparison is also the survey’s.

The statistic is always the one measured over the window’s whole field. 157 windows admit integrations lying inside the primary beam that a narrower field would discard, which recovers a median 2.1 times the on-source time and moves the search statistic by a median 0.07; the whole-field value is the one adopted, on 8 rows of Table 9, and a window’s control ensemble is measured with it, so neither side of the comparison mixes two fields. Where the deposited catalogue’s disposition column disagrees — on 4 crossings that fall below the trigger, and the species list moves none into attribution — the ledger is the authority.

Execution-block quality

The comparison with the control positions assumes each block’s ensemble behaves like noise, and one block makes that false: every one of its 512 probes exceeds the trigger, the smallest at 6.6. A criterion for the condition is therefore applied to every block rather than to the one that prompted it.

Two criteria are built from that statistic and they are not the same test. The statistic is the median of a window’s own 512-position control ensemble, which separates a field with a source in it from a field that is no null at all. A *window* is flagged when that median exceeds twice the median of the same quantity over the survey, 5.5 against 2.74; the flag is a property of the window and cuts across attribution by design, firing on 5 of the 1651 windows in 2 blocks toward 2 stars (BD+05 1668 and HD 48370), and every one of those windows carries a crossing, attributed and unattributed alike. A *block* is judged on the same median divided by the median over the windows of its own class, since a maximum over a larger channel×drift grid sits higher and one threshold should mean the same thing in both; a block fails when the median of that ratio over its windows exceeds twice the scale of its class. The factor is the same in both. Neither was fixed in advance: the condition was noticed in one block and the criterion written afterwards, so it is post hoc, and what makes it trustworthy is that it does not depend on where the line was drawn — the failing block sits at 4.1 and the next worst at 1.12, so any threshold between them selects the same single block.

Applied to all 404 blocks of the primary census, where the median block sits at 1.00, the criterion fails 1: A002_Xc7fa6f_X28a8 toward BD+05 1668. Nothing else is close, and the unscaled form of the rule fails the same 1 block. Its 4 windows are all coarse-channel and hold the survey’s entire Class B crossing population, so it carries 4 of the 56 threshold crossings — 0 of them outranking all 512 control positions of their own window

— and no Class A statistic here has ever contained it. Those 4 appear in Table 9 with their block named and enter no count elsewhere.

The control ensemble: its calibration, and its resolution

The ring geometry is set in §4.1; its calibration and its resolution are the subject here. Over the 1645 census windows that retain a control vector the stellar ranks are consistent with the one half exchangeability implies — median 0.48 with each star carrying equal weight, 0.47 to 0.51 — in both array strata, the compact-array windows measured on each observation’s own antenna table rather than on a 12m beam. That is a weak statement and not a strong one, since a shift smaller than about 0.042 could not have been seen in a sample of this size. The reserved blocks lean further from one half, and Appendix B.1 compares the two samples. The expectations above are summed from the exceedance rate measured at each window’s control positions, so they do rest on that transfer and carry its measured accuracy.

The rank is not a significance test, and the reason is structural. Ranking the star against 512 control positions of the same field decides which windows deserve examination, but $1/(512+1) = 1.95 \times 10^{-3}$ is the *resolution* of the ensemble — the smallest rank it can express — and not the probability that noise produced the window. A survey of 1651 windows would need a rank 64 times finer, and correlation makes it coarser still: controls inside the synthesised beam are not independent, so an annulus resolves no better than 0.023–0.032 however many probes are drawn into it, with the primary beam bounding how many a single pointing contains. That floor is the design’s central limitation: no single window here can be significant on the rank alone, which is why confirmation requires independent recurrence.

The trials budget

The searched cells counted in §4.2 are not independent trials: channel smoothing and a drift grid stepped twice per channel correlate neighbouring ones, which alone would leave 0.19 of them independent. The control ensembles say otherwise, their maxima sitting at 0.995 of the prediction for the maximum of the full searched count treated as independent and Gaussian — a budget predicting 476 windows with a control above T_{trig} against the 455 observed — while fitting each window’s effective count against its own ensemble returns 0.81 of its searched cells. Nothing turns on which is right, because the expectation above is summed from measured rates and not from a cell count: by the cell-count route the number reaching the trigger is 36.5 rather than 36.1, 3 per cent away.

The on-star crossings do exceed that budget, and the attribution accounts for it. A Gaussian tail over every searched cell predicts 37.8 stellar exceedances against the 56 found, but 18 are identified with a catalogued transition, 4 come from the block of §D.2 and 2 fall outside the released windows, leaving 32 against the 29.0 predicted. No excess survives the attribution.

One stratum departs from uniformity, and the departure is where the line emission is. Counting the windows whose stellar peak both reaches the trigger and exceeds all 512 control positions of its own window, the 402 fine-channel windows hold 12 against 0.78 expected, while the

coarse stratum, three times larger and predicting 2.43, contains 0. A coarse channel dilutes a narrow circumstellar line, so the class that can resolve one is the class that finds it — and an unresolved carrier would appear in Class A first for the same reason. Reaching the trigger is holding a crossing, so each of those windows holds one and 10 hold a crossing the mask attributes: the departure is a statement about the windows carrying line emission and not about the fine stratum at large. Drop the trigger condition and the star leads its own ring in 2 further windows, at 4.20 and 4.36, giving 14 over all 1651 windows against 3.2 expected. One quantity, two conditions; the chance comparison uses the conditioned one.

What is left is bounded, and the bounds are null. The real stellar ranks agree at $p = 0.38$ with the 220672 ranks each control takes against the other probes of its own window; radial suppression is below 10 per cent in the mean; and scrambling a window so nothing coherent survives along a drift track carries 4 past the rank floor and changes no disposition.

RADIO-FREQUENCY INTERFERENCE SCREENING

Interference is the first alternative explanation for an unattributed crossing, so the screening this survey has, and the coverage it does not have, are both set out here. Millimetre interferometry is an unusually poor environment for terrestrial interference, though not an empty one: ITU Radio Regulations Article 5 (International Telecommunication Union 2024) carries active allocations throughout 90–275 GHz and WRC-19 identified parts of 275–450 GHz for land-mobile and fixed use. The specific hazard is the 94.0–94.1 GHz Earth exploration-satellite allocation used by spaceborne cloud radar (Stephens et al. 2002; Wehr et al. 2023), a documented contaminant of ALMA Band 3; 1 of the 1651 census windows overlaps it, carrying 0 threshold crossings and 0 rank-passing events, so no result here rests on it. What favours this band is that terrestrial transmitters are sparse compared with centimetre wavelengths and, decisively, that an interferometer is not a total-power detector: a signal not in the far field at the phase centre does not fringe-track but acquires a rapidly varying geometric phase across the array, and is suppressed in the correlation rather than imaged.

The coverage of the screening is uneven, and that is stated rather than smoothed over. The allocation test above and the sky- and intermediate-frequency tests below are computed for every searched window from the released catalogue, as is the 512-position rank screen, which asks whether an excess is at the star or everywhere in the field. Recurrence reaches only the crossings with a covering repeat block, and the visibility fit is a consistency check rather than a filter. The correlator flagging is the observatory’s own, replayed from each block’s delivered calibration script, but it is not recorded per window in the release. *No per-channel interference flag of our own exists, and none can be recovered.* Three measurements follow, and each is a test the population could have failed.

First, clustering in sky frequency. If interference dominated the crossing population, the crossings would pile up at particular sky frequencies. They do cluster — but

TABLE 9

THE CROSSING LEDGER, FROM WHICH EVERY DISPOSITION IN THIS PAPER CAN BE REPRODUCED. ALL 56 THRESHOLD CROSSINGS, INCLUDING THE 4 IN THE FAILING BLOCK OF §5.3.2: THE STAR, ITS EXECUTION BLOCK, THE ALMA BAND, THE CROSSING FREQUENCY AS THE CORRELATOR DELIVERED IT AND THE SAME FREQUENCY IN THE STAR'S REST FRAME, THE SEARCH STATISTIC T_* , THE LARGEST STATISTIC T_{ctrl} AMONG THE WINDOW'S 512 CONTROL POSITIONS, THE NEAREST MASKED TRANSITION AND THE STELLAR-FRAME OFFSET $\Delta\nu_*$ FROM IT, THEN THE RECURRENCE TEST: n_{rep} COVERING REPEAT BLOCKS SPANNING Δt DAYS, THE STATISTIC T_{pers} A CARRIER PERSISTING AT THE DISCOVERY FLUX WOULD HAVE PRODUCED AT THE PREDICTED STELLAR-FRAME CELL, THE STATISTIC T_{rep} MEASURED THERE AT THE CARRIED DRIFT, AND THE EXCLUSION OF EQ. B1, WHICH THREE OF THESE COLUMNS REPRODUCE, TWO OF THEM ALSO GIVING THE LIMIT DEFINED IN APPENDIX B. WHERE THE EMISSION IS PRESENT AGAIN IN A COVERING BLOCK THE TEST RETURNS A RECOVERY AND NOT AN EXCLUSION, SO BOTH T_{rep} AND σ_{excl} ARE BLANK AND THE DISPOSITION SAYS WHICH IT WAS. T_* AND T_{ctrl} COME FROM THE SAME MEASUREMENT OF THE SAME WINDOW; T_{ctrl} GATES NOTHING AND SAYS ONLY WHICH CROSSING WAS EXAMINED FIRST, AND THE DRIFT-MAXIMISED READING OF THE CELL IS IN TABLE 5 FOR THE TWO LEADING EVENTS AND IN THE DATA RELEASE FOR THE REST. ROWS WHOSE WINDOW IS DATA-QUALITY FLAGGED (§D.2) ARE MARKED, AND ON THOSE T_{pers} AND THE EXCLUSION ARE WITHHELD: BOTH ARE BUILT FROM THE DISCOVERY AMPLITUDE, AND IN A WINDOW WHOSE CONTROL RING STANDS AS HIGH AS ITS STAR THAT AMPLITUDE CANNOT BE ASSIGNED TO A POINT SOURCE AT THE STAR. 1 OF THOSE ROWS LIES OUTSIDE THE FAILING BLOCK — HD 48370, A RESOLVED DISC — AND IS WITHHELD FOR THAT REASON AND NOT FOR ITS BLOCK. A CROSSING IS ATTRIBUTED WHERE $|\Delta\nu_*| \leq 50 \text{ km s}^{-1}$ AND THE COINCIDING SPECIES IS ONE SUCH A SYSTEM COULD PLAUSIBLY EMIT; A DAGGER MARKS THE 3 ROWS THAT SATISFY THE FIRST CONDITION AND NOT THE SECOND, FOR THE REASON GIVEN IN APPENDIX C. THE TRANSITION NAMED IS THE NEAREST IN THAT FRAME, SO FAR FROM EVERY TRANSITION IT IS THE MAGNITUDE AND NOT THE NAME THAT CARRIES INFORMATION. BLOCK IDENTIFIERS OMIT THE CONSTANT A002_ PREFIX.

Star	block	$B \nu_{\text{topo}}$ (GHz)	ν_* (GHz)	$T_* T_{\text{ctrl}}$ line	$\Delta\nu_* n_{\text{rep}}$	Δt (d)	T_{pers}	T_{rep}	σ_{excl}	disposition
BD+05 1668	Xc7fa6f_X28a8	6	240.1016	240.1054 73.398 74.69 CS(5-4)	-5911.9	29	0.9-113	n/a+0.15	n/a	unattributed; flagged
BD+05 1668	Xc7fa6f_X28a8	6	224.7422	224.7458 48.303 83.74 H ₂ CO(3 _{1,2} -2 _{1,1})	-1264.5	29	0.9-113	n/a+0.37	n/a	unattributed; flagged
BD+05 1668	Xc7fa6f_X28a8	6	226.7422	226.7458 41.355 75.48 CN(2 _{3/2} -1 _{1/2})	+88.0	29	0.9-113	n/a+0.66	n/a	unattributed; flagged
BD+05 1668	Xc7fa6f_X28a8	6	242.1953	242.1992 39.586 83.29 CS(5-4)	-3349.2	29	0.9-113	n/a+0.77	n/a	unattributed; flagged
β Pic	Xf5d76d_Xeb8	3	115.2603	115.2711 30.765 7.34 CO(1-0)	-0.2	4	0.0-53	n/a	—	n/a attributed; recurs
β Pic	Xf5d76d_Xe19	3	115.2603	115.2711 27.684 6.48 CO(1-0)	-0.2	4	0.0-53	n/a	—	n/a attributed; recurs
HD 48370	Xc26103_X155a	6	230.5117	230.5243 26.834 18.36 CO(2-1)	-17.8	—	—	n/a	—	n/a attributed; flagged
β Pic	Xf4df6f_Xc7	3	115.2620	115.2711 17.907 5.76 CO(1-0)	-0.2	4	53	n/a	—	n/a attributed; recurs
β Pic	Xf5d76d_Xcc5	3	115.2604	115.2711 17.288 5.52 CO(1-0)	-0.2	4	0.0-53	n/a	—	n/a attributed; recurs
β Pic	Xf5d76d_X32f1	3	115.2606	115.2713 14.654 7.27 CO(1-0)	+0.4	4	0.0-53	n/a	—	n/a attributed; recurs
β Pic	Xd9668b_X3a90	6	230.5161	230.5378 11.681 9.12 CO(2-1)	-0.3	5	1.0-1983	n/a	—	n/a attributed; recurs
β Pic	X6f1341_X1484	6	230.5284	230.5379 9.832 8.20 CO(2-1)	-0.1	5	25-1984	n/a	—	n/a attributed; recurs
β Pic	X7116f1_X1785	6	230.5272	230.5384 9.677 14.13 CO(2-1)	+0.5	5	25-1959	n/a	—	n/a attributed; recurs
β Pic	Xd9668b_Xa9df	6	230.5160	230.5377 8.948 8.41 CO(2-1)	-0.4	5	1.0-1984	n/a	—	n/a attributed; recurs
β Pic	Xa7a216_X23e4	6	230.5273	230.5377 7.988 6.03 CO(2-1)	-0.4	5	601-1313	n/a	—	n/a attributed; recurs
61 Vir	Xc079b5_X82f	7	346.3138	346.3232 5.958 5.65 SO(9 ₈ -8 ₇)	-177.6	3	0.1-3.0	9.7+0.39	4.9	unattributed; no recurrence
HD 285968	Xd98580_X3bf6	6	230.5021	230.5447 5.956 8.12 CO(2-1)	+8.7	3	13-44	n/a	—	n/a attributed; recurs
HD 285968	Xdb7ab7_X5b4e	6	230.5116	230.5447 5.868 7.22 CO(2-1)	+8.7	3	5.0-44	n/a	—	n/a attributed; recurs
HD 285968	Xdb4b9a_X108f	6	230.5100	230.5447 5.835 7.94 CO(2-1)	+8.7	3	5.0-39	n/a	—	n/a attributed; recurs
CP-72 2713	Xff0235_X4a6d	7	344.2697	344.2965 5.809 5.68 SO(8 ₈ -7 ₇)	-12.3 [†]	5	0.1-2.1	6.3-0.82	4.8	unattributed; no recurrence
HD 139084 B	Xcd8029_Xb6b0	7	345.8241	345.8241 5.763 17.70 CO(3-2)	+24.4	—	—	—	—	—
HD 14055	Xff3d8_Xd46	7	331.7699	331.7664 5.575 5.99 ¹³ CO(3-2)	+1068.7	30	0.9-650	170.4-0.02	5.6	unattributed; no recurrence
HD 23484	X10b22d7_X2445	6	230.7904	230.8006 5.520 5.55 CO(2-1)	+341.5	4	0.1-2.1	10.4+1.25	4.3	unattributed; no recurrence
HD 161868	Xff45a6_X10ca8	7	345.6338	345.6491 5.481 6.16 SO(2 ₃ -2 ₁)	-48.0 [†]	13	0.0-6.1	92.2+0.24	5.5	unattributed; no recurrence
HIP 10679	Xa95c04_X14cd	6	220.4185	220.4035 5.456 6.76 ¹³ CO(2-1)	+6.5	—	—	—	—	—
HD 14055	X11adad7_X25c5e7	7	345.4564	345.4303 5.448 6.82 SO(2 ₃ -2 ₁)	-237.8	30	0.1-653	14.0+0.03	5.1	unattributed; no recurrence
HD 207129	Xc19d6f_X5706	6	230.4226	230.4070 5.433 5.52 CO(2-1)	-170.3	4	0.8-425	69.7-0.93	5.5	unattributed; no recurrence
β Pic	Xd9668b_X3a90	6	241.7512	241.7739 5.420 5.81 CS(5-4)	-3869.7	1	1.0	5.7+0.72	3.4	unattributed; no recurrence

TABLE 10

The crossing ledger, continued. THE REMAINING 28 OF THE 56 THRESHOLD CROSSINGS, IN THE SAME ORDER AND WITH THE SAME COLUMNS AS TABLE 9, WHOSE CAPTION DEFINES THEM.

Star	block	$B \nu_{\text{topo}}$ (GHz)	ν_* (GHz)	$T_* T_{\text{ctrl}}$ line	$\Delta\nu_* n_{\text{rep}}$	Δt (d)	T_{pers}	T_{rep}	σ_{excl}	disposition
HR 1010	Xc6fa08_X1053	6	230.5133	230.5287 5.410 5.65 CO(2-1)	-12.0	7	2.0-14	60.1<+4.60	>5.0	attributed; not recovered
ALMA J153702653-33192492	Xff0235_X8392	6	230.5222	230.5392 5.405 13.41 CO(2-1)	+1.6	6	18-141	6.2<+4.15	>1.4	attributed; not recovered
HD 110411	Xe45e29_Xb7d4	6	219.4538	219.4357 5.393 5.85 C ¹⁸ O(2-1)	-170.2	1	10	6.7	+0.62	3.8 unattributed; no recurrence
HD 31392	X129754b_X6946	6	231.1998	231.2231 5.289 5.81 H(30a)	-876.3	8	0.1-41	15.7	-0.26	5.1 unattributed; no recurrence
HD 14055	Xff99e1_X24be0	7	331.7078	331.7035 5.277 5.69 ¹³ CO(3-2)	+1011.6	30	0.9-652	252.2	+0.15	5.3 unattributed; no recurrence
AU Mic	Xa42f75_X319	6	230.6864	230.6705 5.274 6.07 CO(2-1)	+172.3	2	310-455	6.4	-0.16	4.2 unattributed; no recurrence
HD 69830	X12209f7_X16edb7	7	322.4325	322.4444 5.259 5.42 C ¹⁸ O(3-2)	-6268.5	5	0.1-29	8.9	-0.48	4.8 unattributed; no recurrence
ALMA J153702653-33192492	Xfe62c1_X398	6	216.1844	216.2036 5.228 6.02 SiO(5-4)	-1244.6	7	18-126	11.0	+1.03	4.3 unattributed; no recurrence
AU Mic	X899169_X1838	6	230.8252	230.8294 5.221 5.85 CO(2-1)	+379.0	2	145-310	10.0	+0.19	4.5 unattributed; no recurrence
HD 14055	Xff99e1_X2c3bb	7	345.5256	345.5216 5.186 5.66 SO(2 ₃ -2 ₁)	-158.6	30	0.1-651	172.8	+2.00	5.1 unattributed; no recurrence
HD 14055	Xffde1_X3835	7	345.0099	345.0070 5.177 5.66 SO(2 ₃ -2 ₁)	-604.8	30	0.1-649	187.2	-1.78	5.2 unattributed; no recurrence
HD 14055	Xff99e1_X2b2ee	7	344.7007	344.6965 5.175 5.92 SO(8 ₈ -7 ₇)	+336.0	30	0.1-651	210.7	-0.00	5.2 unattributed; no recurrence
LHS 1140	Xd9d7c7_Xbf02	6	230.6355	230.6268 5.136 5.68 CO(2-1)	+115.5	4	5.0-27	7.3	+0.33	4.0 unattributed; no recurrence
HD 10647	Xff294f_Xaf70	7	331.6570	331.6943 5.120 5.91 ¹³ CO(3-2)	+1003.3	10	0.1-592	125.8	+0.39	5.1 unattributed; no recurrence
CD-57 1054	Xc525f_X5aa4	7	346.5099	346.5265 5.108 5.71 SO(9 ₈ -8 ₇)	-1.7 [†]	1	39	6.0	-0.27	4.1 unattributed; no recurrence
ALMA J153702653-33192492	Xf8f6a9_X10359	6	217.7292	217.7269 5.091 5.41 H ₂ CO(3 _{0,3} -2 _{0,2})	-680.5	7	3.8-242	11.6	+0.07	4.6 unattributed; no recurrence
GJ 849	Xd9668b_X841a	6	230.8769	230.8581 5.090 5.83 CO(2-1)	+416.3	5	0.1-51	9.3	-1.69	5.3 unattributed; no recurrence
ALMA J153702653-33192492	Xfe62c1_X398	6	218.8729	218.8923 5.090 5.56 H ₂ CO(3 _{2,1} -2 _{2,0})	+181.2	7	18-126	10.5	-1.04	5.0 unattributed; no recurrence
HD 14055	X1001fdc_X29ae	7	331.1559	331.1545 0.862 6.26 ¹³ CO(3-2)	+513.8	—	—	—	—	—
β Pic	Xd9668b_Xa9df	6	241.8367	241.8594 5.079 5.99 CS(5-4)	-3765.1	1	1.0	4.8	-0.63	3.9 unattributed; no recurrence
ALMA J153702653-33192492	Xf8f6a9_X10359	6	218.6711	218.6688 5.048 5.77 H ₂ CO(3 _{2,1} -2 _{2,0})	-125.1	7	3.8-242	11.6	+0.00	4.6 unattributed; no recurrence
HD 69830	X12277ed_X178c67	7	321.2005	321.2149 5.046 6.13 C ¹⁸ O(3-2)	-7387.7	5	6.0-36	10.2	-0.46	4.7 unattributed; no recurrence
HR 1010	Xc66ce1_Xaa9	6	230.8348	230.8491 5.044 5.72 CO(2-1)	+404.6	7	1.1-14	10.3	-1.53	5.2 unattributed; no recurrence
61 Vir	Xc067f7_X822	7	345.5599	345.5678 5.038 5.79 SO(2 ₃ -2 ₁)	-118.5	3	2.0-3.0	9.3	+0.45	4.2 unattributed; no recurrence
HD 170773	Xf0235_X8c42	7	331.0101	331.0234 5.029 5.75 ¹³ CO(3-2)	+394.8	4	0.9-113	74.8	+1.56	4.9 unattributed; no recurrence
ALMA J153702653-33192492	Xd9d7c7_Xbb64	7	347.2179	347.1890 5.009 5.68 SiO(8-7)	-122.2	1	0.1	5.0	-2.06	5.0 unattributed; no recurrence
HD 14055	X1190de8_Xd526	7	344.4676	344.4515 5.006 5.96 SO(8 ₈ -7 ₇)	+122.7	30	20-611	11.6	-0.60	4.8 unattributed; no recurrence
η Crv	X122b6ff_X1041e	7	357.2815	357.2475 5.006 5.50 HCO ⁺ (4-3)	+431.3	1	87	11.0	-0.62	4.8 unattributed; no recurrence

on the survey’s own molecular lines. The test needs a measured peak frequency in the frame in which an interfering carrier would stand still, so it runs topocentrically over the 51 of the 56 crossings for which the release records one, against the transitions the search was tuned to. For the other 5, all β Pic carbon-monoxide crossings, the release gives the velocity offset from the line instead, and a frequency reconstructed from a line offset cannot be used to ask whether frequencies crowd onto lines. Of the 51, 11 fall within $\pm 50 \text{ km s}^{-1}$ of a searched transition against 4.7 expected if each window’s peak fell anywhere on that window’s own channel grid, $p = 0.0007$. The densest band, 230.422–230.723 GHz, holds 12 of them, 9 within that velocity window of CO(2 $\rightarrow$ 1); and once the line-attributed crossings are set aside, the 40 that remain show no clustering at all ($p = 0.111$). *The concentration is the attribution, not a residue left over after it.* Table 9 counts the same band as 14 crossings with 11 attributed, a different test over a different set — all 56, each attributed in its own star’s rest frame — and the 2 it adds are the ones whose peak frequency this test has no measurement of.

Second, fixed intermediate frequency. An artefact of the signal chain would sit at a fixed position within the spectral window rather than at a fixed sky frequency. This test is run over every window for which the catalogue releases a peak frequency — 1397 of the 1651, of which only 51 carry a threshold crossing — because such an artefact would occupy the same intermediate frequency in all of them and not only where the trigger fires. It does not: the peaks’ fractional positions within their own windows have median 0.50 and are uniform.

Third, a cross-target occupancy screen over every window the pipeline extracted, and not only the crossings. That set is wider than the census: 3027 windows toward 133 stars, being the census together with the rows its audit removed, the reserved hold-out, the archival extension, the out-of-sample control, and 64 windows in 16 blocks that entered none of them. Interference belongs to the observatory and not to the star, so a terrestrial carrier should recur at the same sky frequency toward unrelated stars. It does not even for the leading event: the one unattributed rank-passing event has a released crossing frequency; 40 further windows, toward 24 other stars in 32 other blocks, cover that frequency, and none of them carries a crossing in the same channel. With each window’s peak randomised on its own channel grid, the number of cross-star coincidences within 5 MHz is 569 against 554 ± 13 expected ($p = 0.15$); among the 109 windows whose peak exceeds the trigger it is 4 against 5.2 ± 1.9 , i.e. below the null, and with the molecular-line mask applied 1 against 2.1 ± 1.3 . There is no excess to explain. The mask has to be applied first, because 35 of those 109 windows sit on a masked transition (CO(2–1) $\times$ 17, CO(1–0) $\times$ 10, CO(3–2) $\times$ 4, ^{13}CO (2–1) $\times$ 3, SiO(5–4) $\times$ 1), and a carbon monoxide line falls at nearly the

same sky frequency toward every nearby star — exactly the coincidence the screen is looking for.

One execution block fails these tests, and it fails them as a block rather than as a carrier. Its 4 crossings toward BD+05 1668 are the highest statistics in the survey outside β Pictoris and HD 48370, and the field statistics that identify the failure are in §5.3.2. What belongs here is whether interference accounts for it, and three readings say not. The visibility fit returns a large imaginary part, placing the emission away from the phase centre. The block’s 24 sibling executions of the same field, with the same array and the same spectral setup, produce no crossing at all. And it is not a carrier at a fixed sky frequency: each of the four frequencies is covered by 127–137 further windows toward 17–22 other stars, and 0 of them carries a crossing in the same channel.

One reading does point at a mechanism, and it can be tested across the survey. 2 of the 4 sit at the identical channel of two different spectral windows, which is the signature of a fixed intermediate frequency — and a feature with no geometric fringe phase, a spur common to every antenna being the obvious case, does not fringe-track, so it gathers at the phase centre rather than at the star. Here the two are a twentieth of an arcsecond apart, so the block cannot test its own hypothesis; the survey can, because the star lies 5.0–8.1 arcsec from the phase centre in 75 windows whose control annulus brackets it, putting a median of 15 probes within two synthesised beams of the phase centre and the star well outside. Against the probes at the same distance from the star but elsewhere in azimuth — matching the radius matters, or a centrally concentrated disc would imitate the effect — those probes are higher by $+0.09 \pm 0.08$ in the statistic, consistent with zero and bounded by 0.26, smaller by a factor of about 33 than the 8.4 by which this block’s ring is raised. *The phase centre is not where the excess lives.* The evidence supports a data-quality failure of one execution block, and that is how these 4 crossings are dispositioned (§5.3.2). Two criteria are involved and they are not the same one (Appendix D.2): the window-level flag, which fires on 5 of the 1651 windows toward 2 stars (BD+05 1668 and HD 48370), the second of them a resolved disc rather than a defect; and the block criterion built from it, which fails 1 block of the 404 searched. It is the block criterion that disposes of these crossings.

None of this is a substitute for the ON/OFF cadence a dedicated search uses, and we do not present it as one. We claim only that *no identified interference mechanism accounts for the unattributed events*: allocation tables cannot exclude harmonics, intermodulation products, unconventional emitters or near-field sources, and archival data without a simultaneous off-source strategy cannot do better. The three tests establish that the crossing population as a whole carries none of the signatures interference leaves, which is a statement about the population and not a clearance of any individual crossing.

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